Artificial Intelligence

Algorithms and Applications with Python

Chapter 11



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Outline

11 Building Productive AI-based Systems

- 11.1 The Challenge of Applied AI
- 11.2 Robotic Agents
- 11.3 Human-AI Interaction
- 11.4 AI Ethics and Governance
- 11.5 How to Continue your AI Journey

Exam Preparation and Course Repetitorium

Capstone Project / Case Challenge



► **What you will learn:**

- Why many AI-based Information systems and intelligent agents fail in practice and how to tackle common problems in implementing AI-based information systems and intelligent agents
- Outlook and Future steps if you are interested in an AI-job

► **Duration:**

- 135 min + FAQs

► **Relevant for Exam:**

- 11.1-11.4

11.1 Current AI Systems Outperform Humans in Many Tasks

- In the last chapters, we have discussed how to build and use AI and its potential for automation in many scenarios.
- But is AI really better than humans in decision-making besides these “academic” examples?
- Yes! There is overhelming evidence that artificial intelligence, clearly outperforms human decision-making:
  - **Paul Meehl**: „Clinical vs. Statistical Prediction: A Theoretical Analysis and a Review of the Evicence“
  - **Daniel Kahneman:** „Thinking Fast and Slow“, or „Noise“
- **But…**

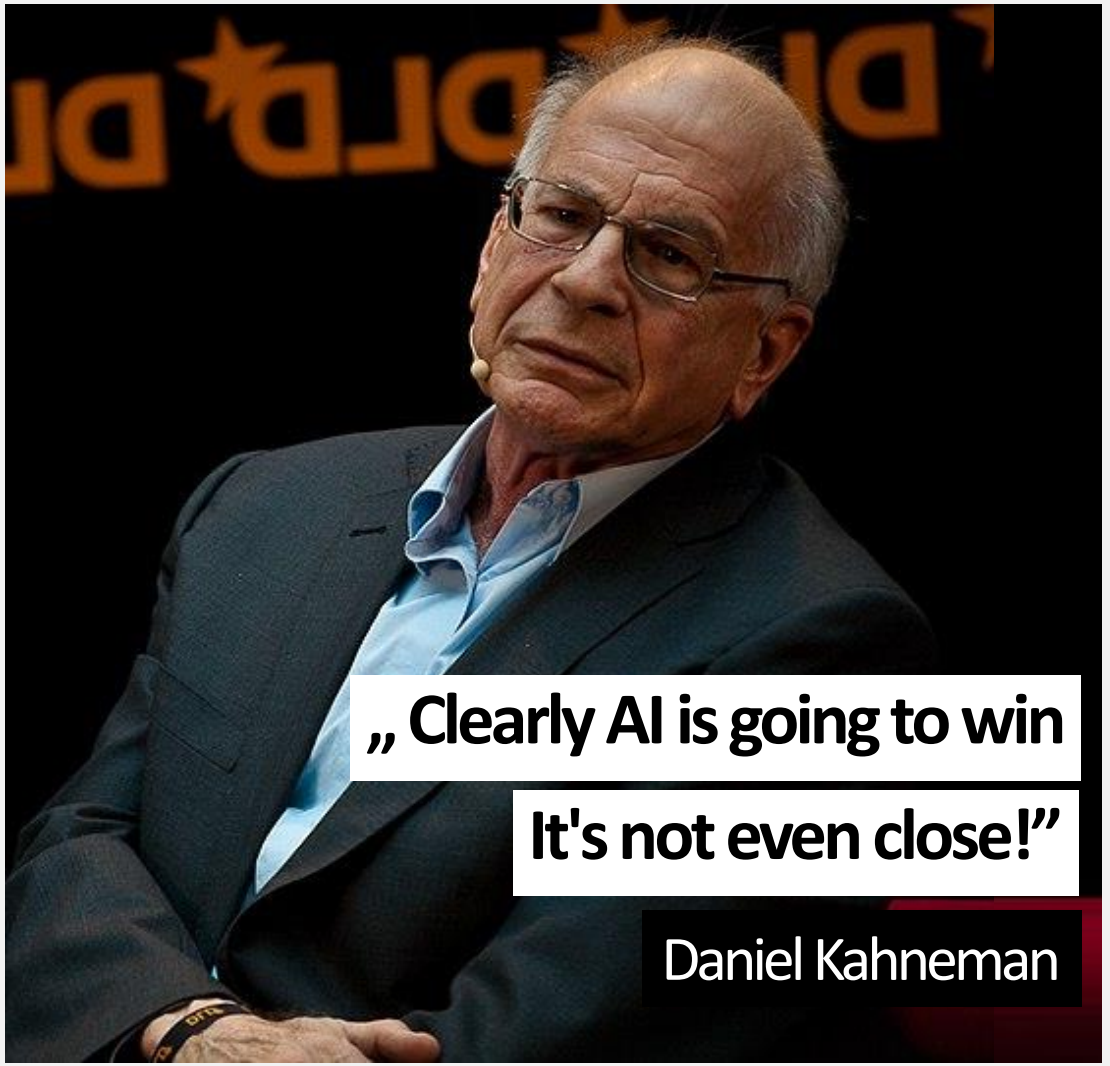


Image source: ↗Daniel Kahneman (2009) by nrkbeta from Wikimedia / ↗CC BY-SA 3.0

11.1 Why Do They Fail?



11.1 Why Do They Fail? ▶ MD Anderson Cancer Center and IBM Watson



IEEE SPECTRUM

Feature | Biomedical | Diagnostics

02 Apr 2019 | 15:00 GMT

How IBM Watson Overpromised and Underdelivered on AI Health Care

After its triumph on *Jeopardy!*, IBM's AI seemed poised to revolutionize medicine. Doctors are still waiting

By **Eliza Strickland**

https://spectrum.ieee.org/biomedical/diagnostics/how-ibm-watson-overpromised-and-underdelivered-on-ai-health-care

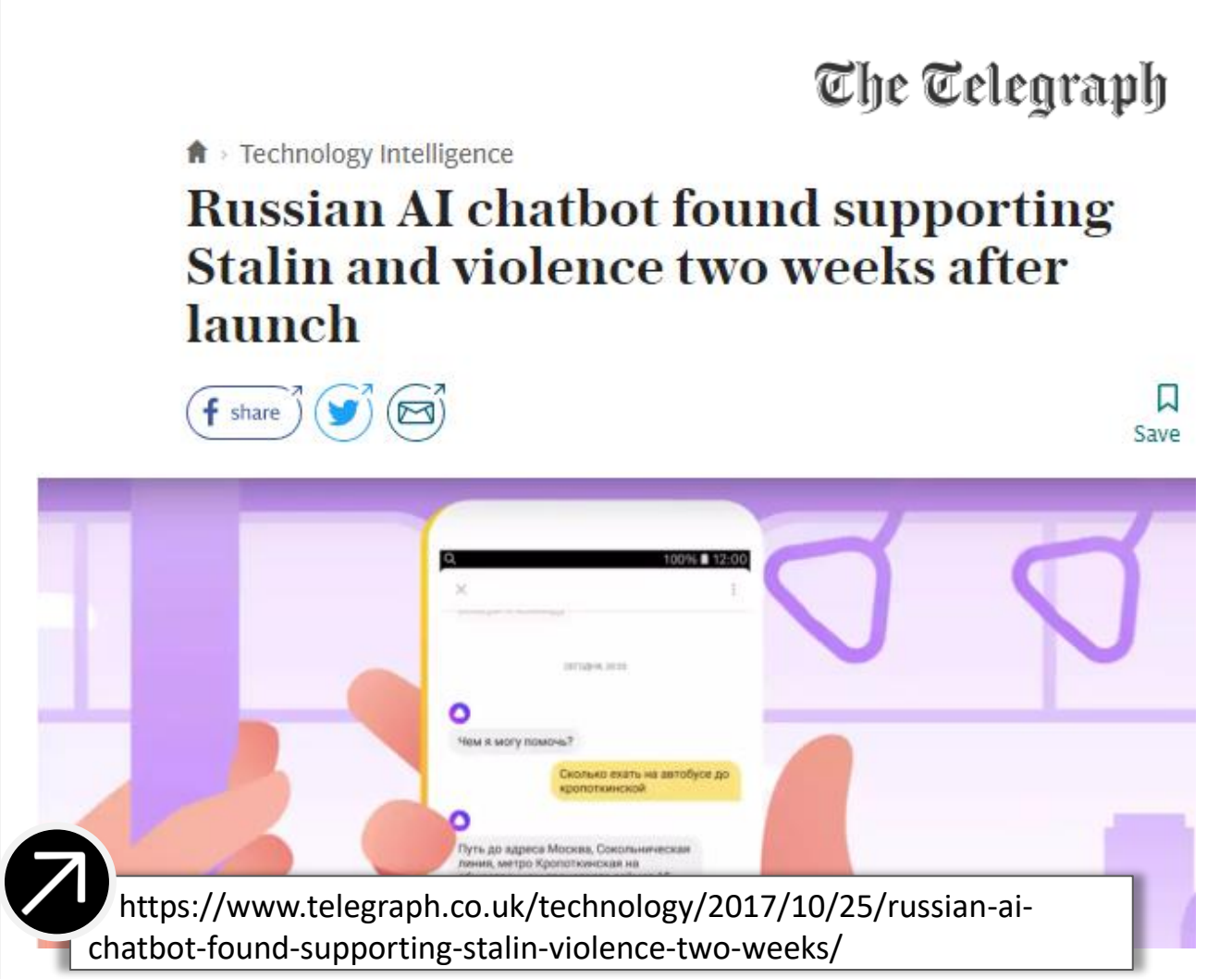
- 62 Million dollars has been spend by University of Texas before the contract was canceled
- Chamath Palihapitiya: "Watson is a joke, just to be honest. I think that IBM is excellent at **using their sales and marketing infrastructure to convince people** who have **asymmetrically less knowledge to pay** for something."

11.1 Why Do They Fail? ▶ DPD Customer Chatbot



- A DPD UK chatbot unexpectedly criticized its own company during a customer interaction, labeling DPD as "the worst delivery company in the world.“
- Following this incident, DPD UK promptly deactivated the chatbot and initiated updates to prevent similar occurrences in the future.

11.1 Why Do They Fail? ▶ Violent Yandex's Alice Chatbot



The Telegraph

Technology Intelligence

Russian AI chatbot found supporting Stalin and violence two weeks after launch

share

Save

https://www.telegraph.co.uk/technology/2017/10/25/russian-ai-chatbot-found-supporting-stalin-violence-two-weeks/

- Users of the "Alice" assistant have reported it responding positively to questions about domestic violence and saying that "enemies of the people" must be shot
- Supports wife-beating, child abuse and suicide

11.1 Why Do They Fail? ▶ Google Map Classification

golem.de
IT-NEWS FÜR PROFIS

GOOGLE MAPS IN BERLIN

Wenn aus Aussetzfahrten eine neue U-Bahn-Linie wird

Kartendienste sind für Touristen wie auch Ortskundige längst eine willkommene Hilfe. Doch manchmal gibt es größere Fehler. In Berlin werden beispielsweise einige Kleinprofil-Linien falsch gerendert. Dabei werden betriebliche Besonderheiten dargestellt.

Von Andreas Sebayang

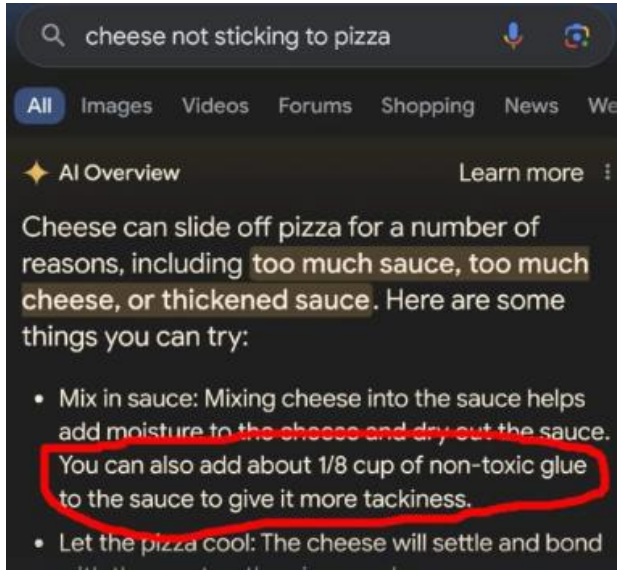
9. Juli 2019, 11:12 Uhr

https://www.golem.de/news/google-maps-in-berlin-wenn-aus-aussetzfahrten-eine-neue-u-bahn-linie-wird-1907-142415.html

- Google determines the routes of tram lines through AI.
- However, this leads sometimes to obviously wrong results: data from public transport was misinterpreted and Google Maps had for some days new underground lines in Berlin.

11.1 Why Do They Fail? ▶ Gemini / LLM Fail





reddit

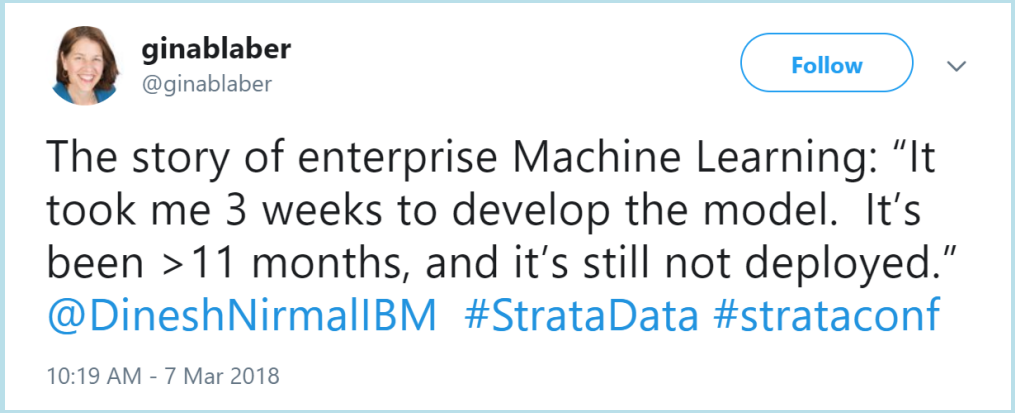
Gemini

- In May 2024, Google's AI-powered search feature, known as "AI Overviews" powered by the Gemini model, faced criticism for providing inaccurate and potentially hazardous advice
- Notably, when users inquired about preventing cheese from sliding off pizza, the AI suggested adding non-toxic glue to the pizza sauce to increase tackiness.
- The erroneous recommendation originated from a satirical comment on Reddit (10 years ago)

11.1 Building Productive AI-based Information Systems is hard…

"60% of models developed to operationalize them were never actually operationalized"

Gartner Data Science Team Survey of January 2018



Why 85% of AI projects fail

TechRepublic

The Myth of Agile AI/Machine Learning in the Enterprise

Eric Broda Follow
Jul 31, 2019 · 13 min read ★

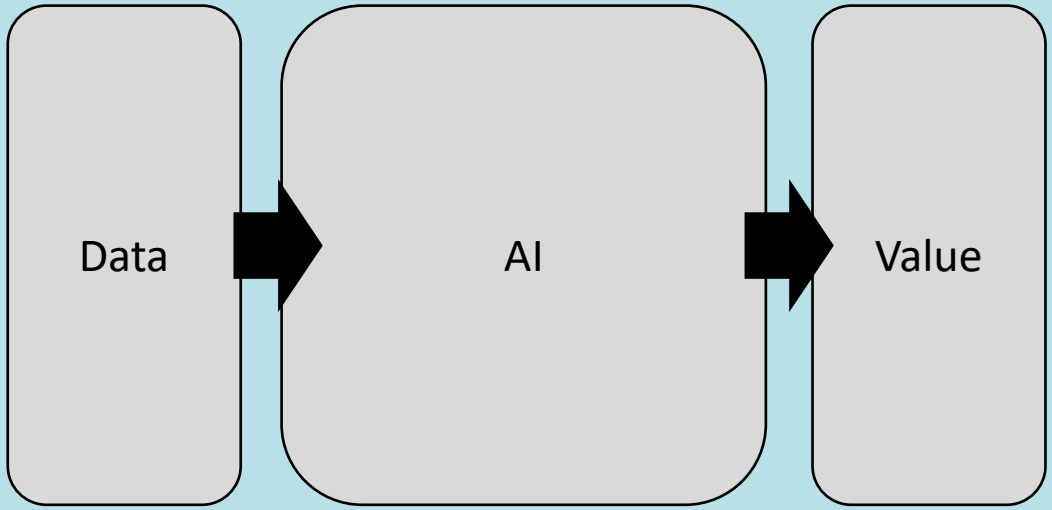
Are AI applications failing to live up to hype in enterprise adoption?

ComputerWeekly.com

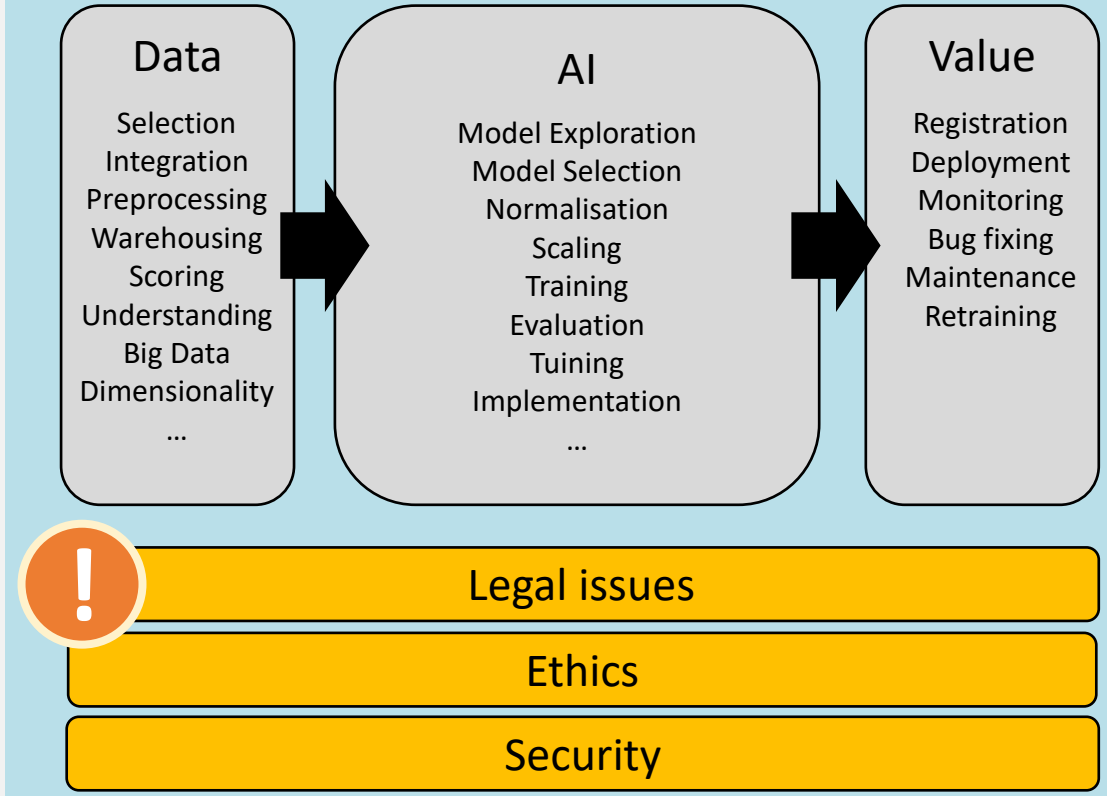
? Why do AI projects fail in real life?

11.1 Problem: Building Productive and Usefull AI Systems is Hard

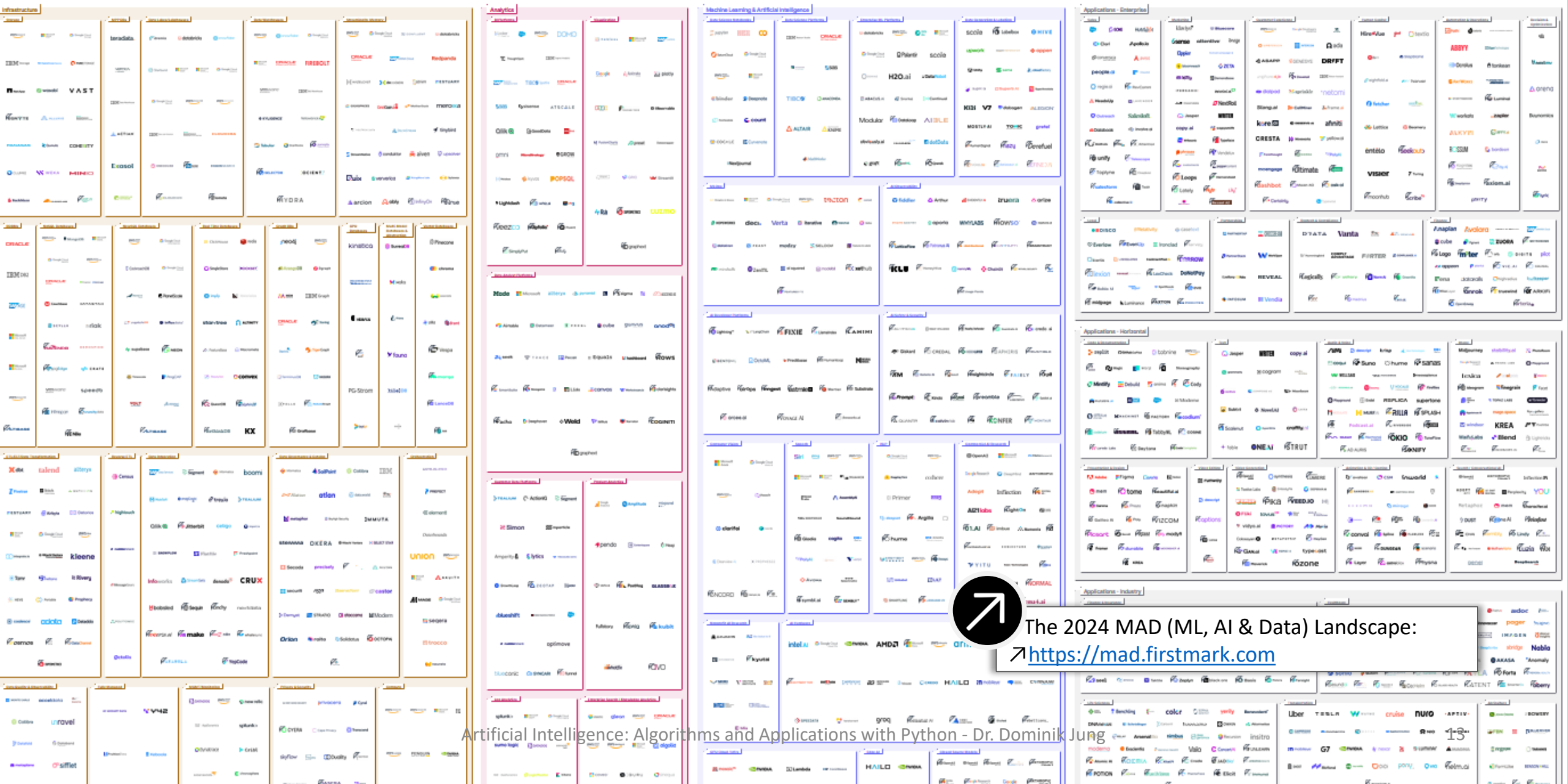
What companies think AI looks like



What it actually is

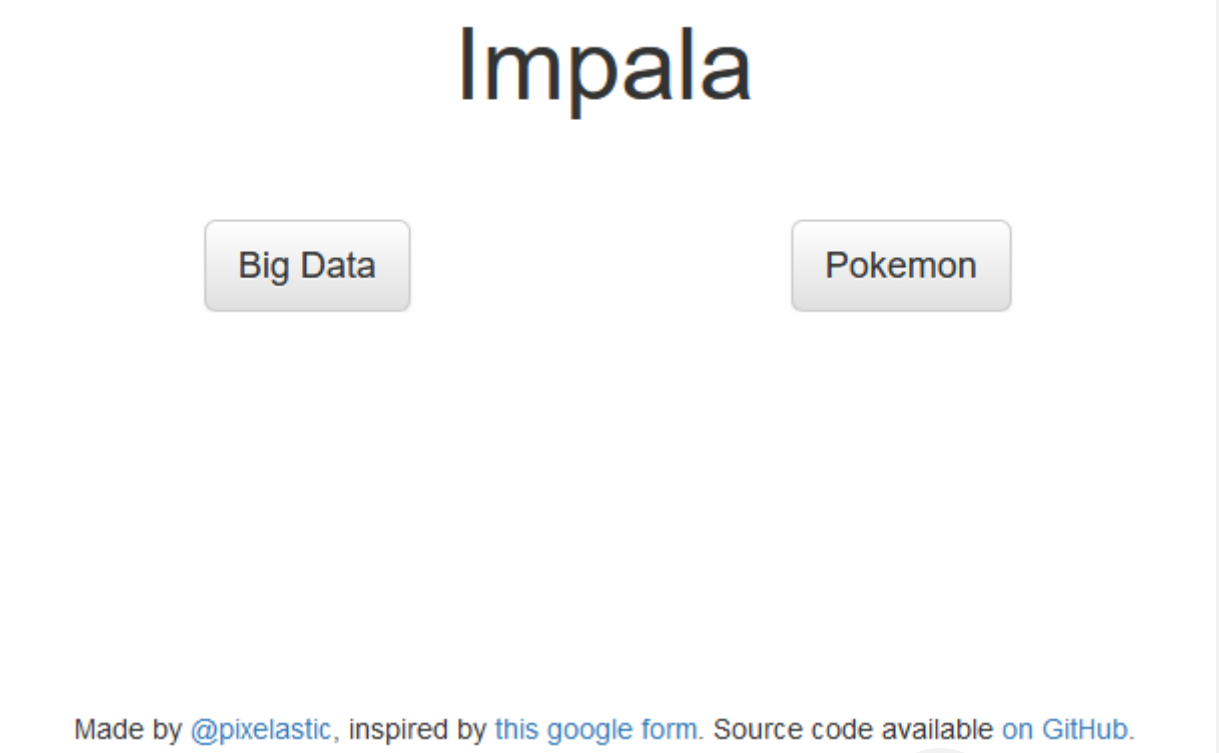


11.1 ML, AI & Data Landscape (2024)



The 2024 MAD (ML, AI & Data) Landscape:
↗https://mad.firstmark.com

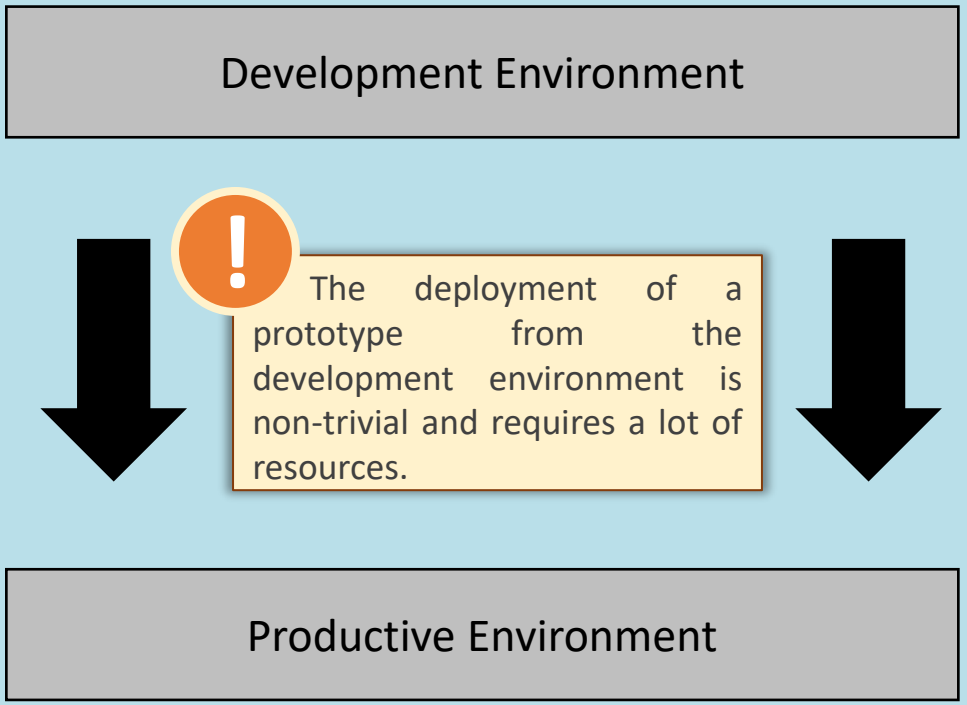
11.1 AI Tool or Pokémon?!



AI Tool or Pokémon Quiz:
↗http://pixelastic.github.io/pokemonorbigdata/

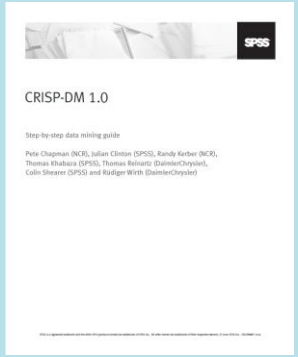
11.1 Lifecycle Approach to Structure AI-based IS Projects

- Most projects consist of several phases and aspects, which together make up an AI-based IS's **life cycle**.
- Spans the inception of an AI use case up until its productive rollout.
- Many popular frameworks to structure projects over the life cycle: CRISP-DM, KDD, TDSP,...
- Best-Practice in AI projects: The frameworks are used in addition to agile or other software development approaches (!).



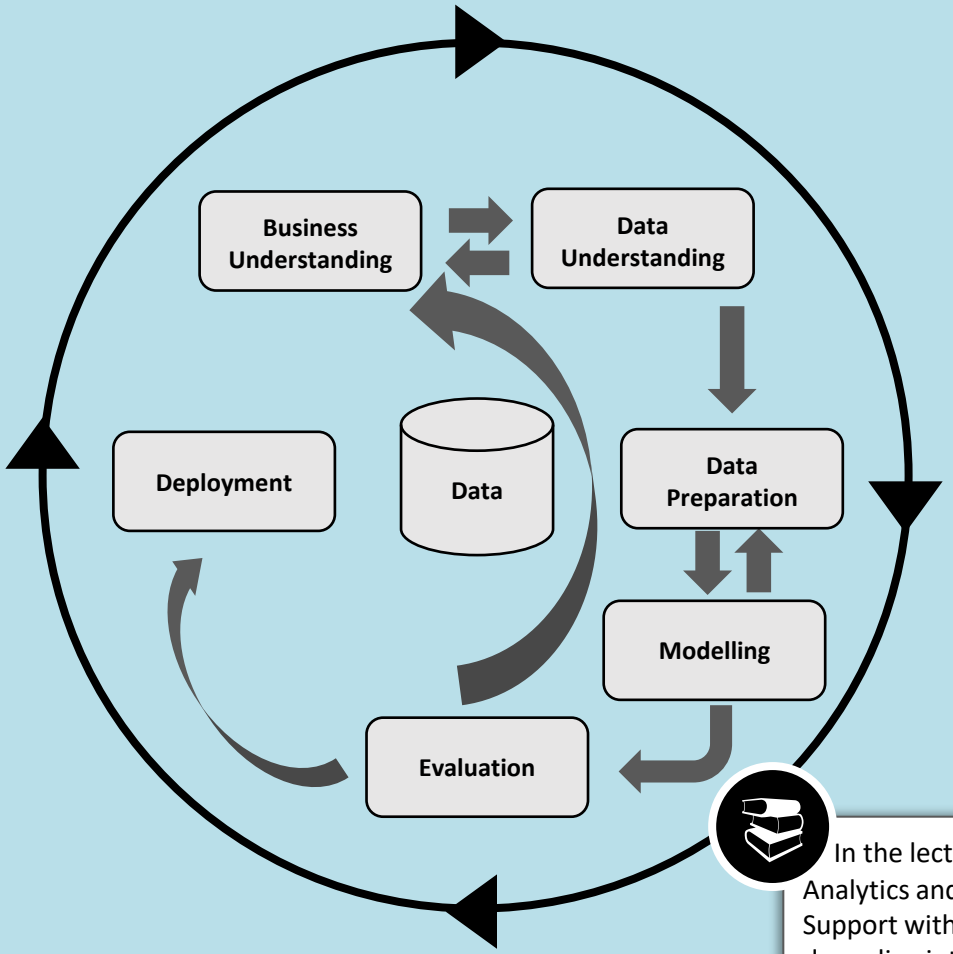
11.1 How will your AI Job Workday look like: CRISP-DM

- Cross-Industry Standard Process for Data Mining (CRISP-DM)
- Process model describing commonly used approaches that data science experts use to tackle problems



Free "CRISP-DM 1.0 *Step-by-step data mining guide*"

In-depth documentation and process guide

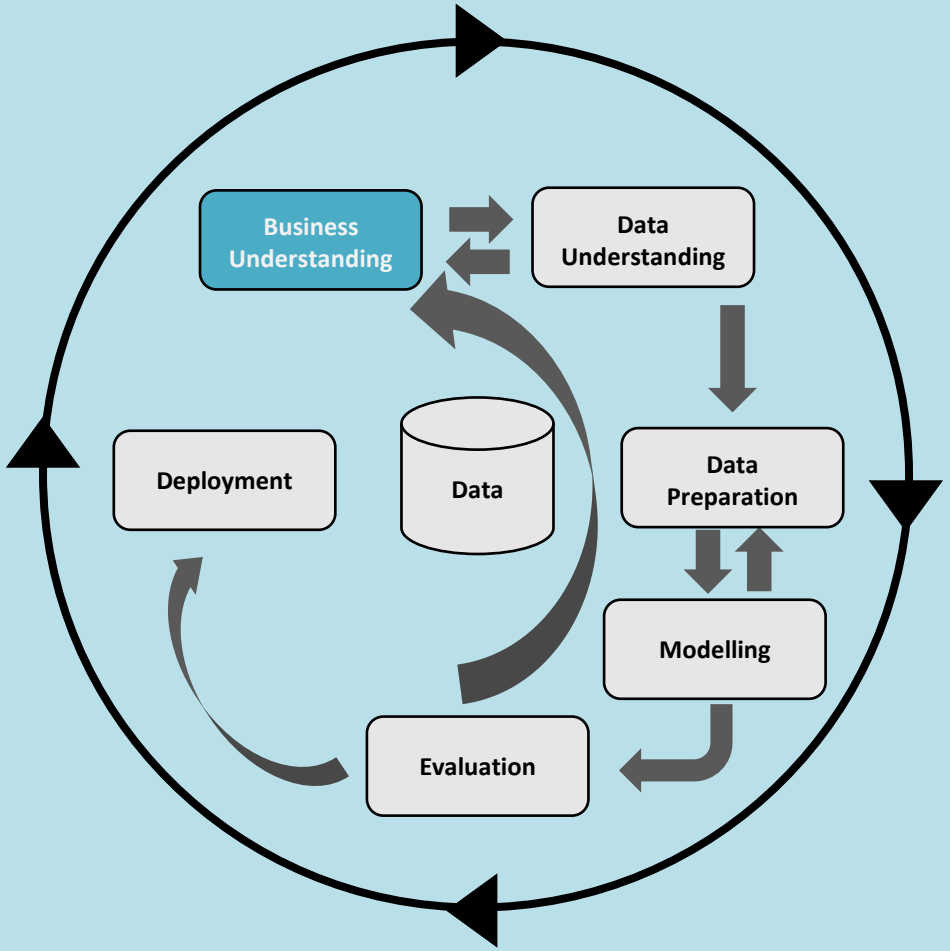


In the lecture "Business Analytics and Decision Support with R" we make a deep dive into the CRISP-DM

Chapman Pete et al. (1999)

11.1 CRISP-DM – Business Understanding

- **What activities are related to the business understanding phase?**
  - Specifying the problem
  - Identifying objectives
  - Understanding of requirements
- **Other common activities:**
  - Make a project plan
  - Define project team
  - Specify project method (e.g. SCRUM)

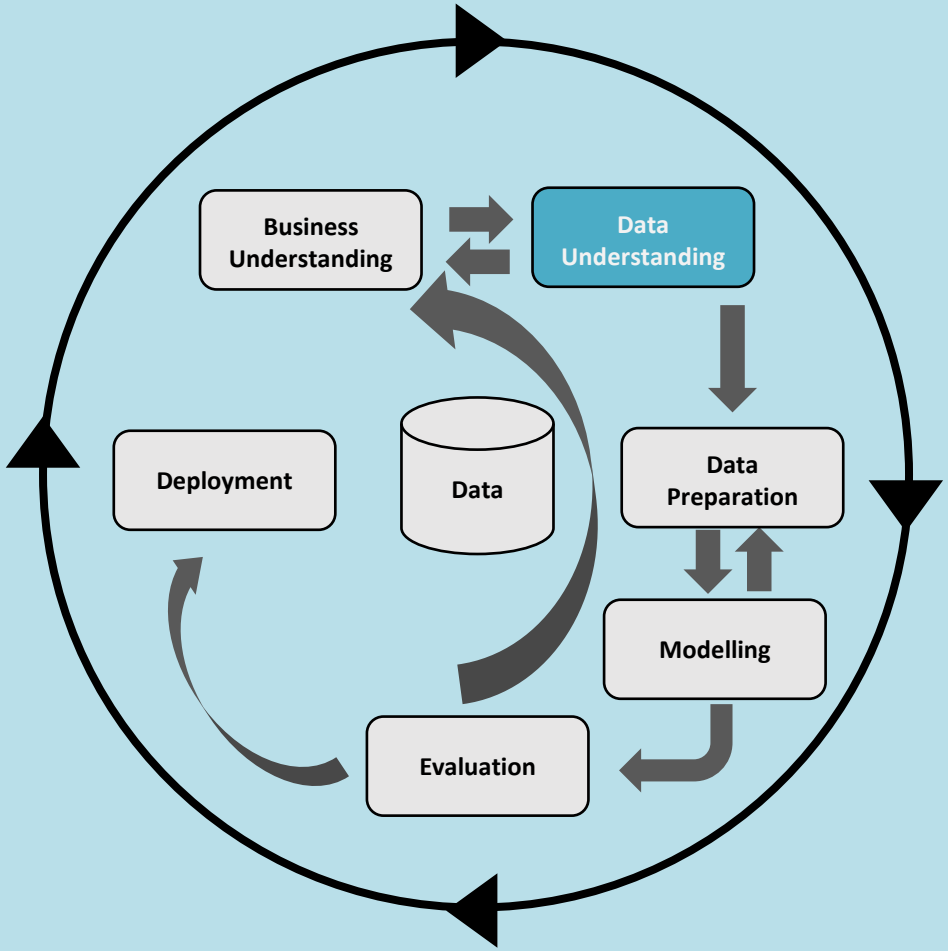


Chapman Pete et al. (1999)

11.1 CRISP-DM – Data Understanding

- **What activities are related to the data understanding phase?**
  - Initial data collection and familiarization
  - Data quality problems identification

- **Other common activities:**
  - Make a data catalogue
  - Define data requirements

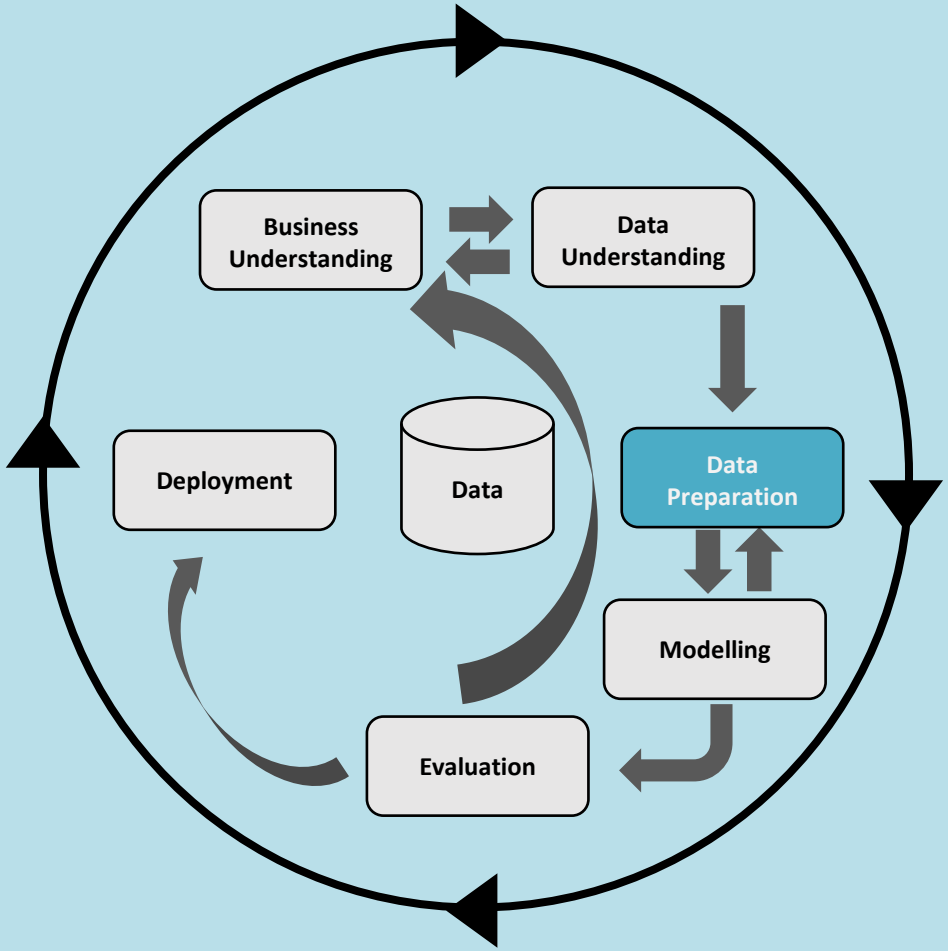


Chapman Pete et al. (1999)

11.1 CRISP-DM – Data Preparation

- **What activities are related to the data preparation phase?**
  - Table, record and attribute selection
  - Data integration
  - Data preprocessing, transformation and cleaning

- **Other common activities:**
  - Define data architecture

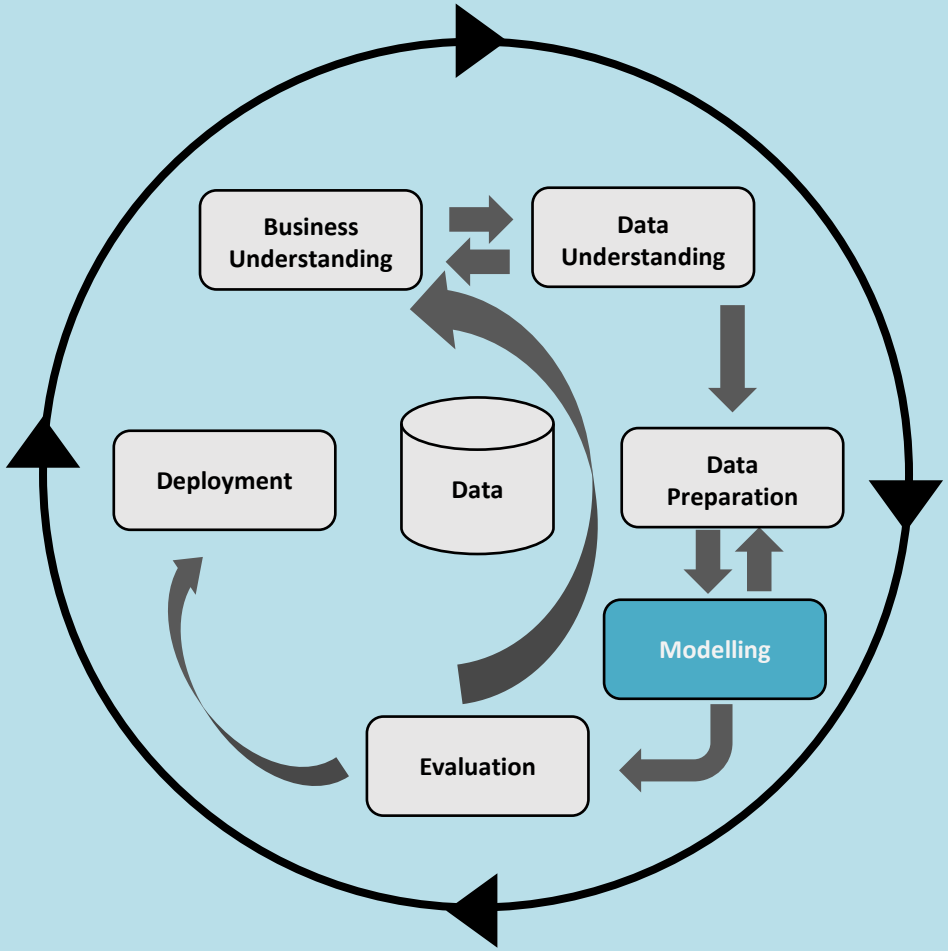


Chapman Pete et al. (1999)

11.1 CRISP-DM – Modelling

- **What activities are related to the modelling phase?**

  - Modeling techniques selection
  - Model application to the problem
  - Parameter calibration of the used algorithms
  - Model assessment
  - Understand the logic behind the model

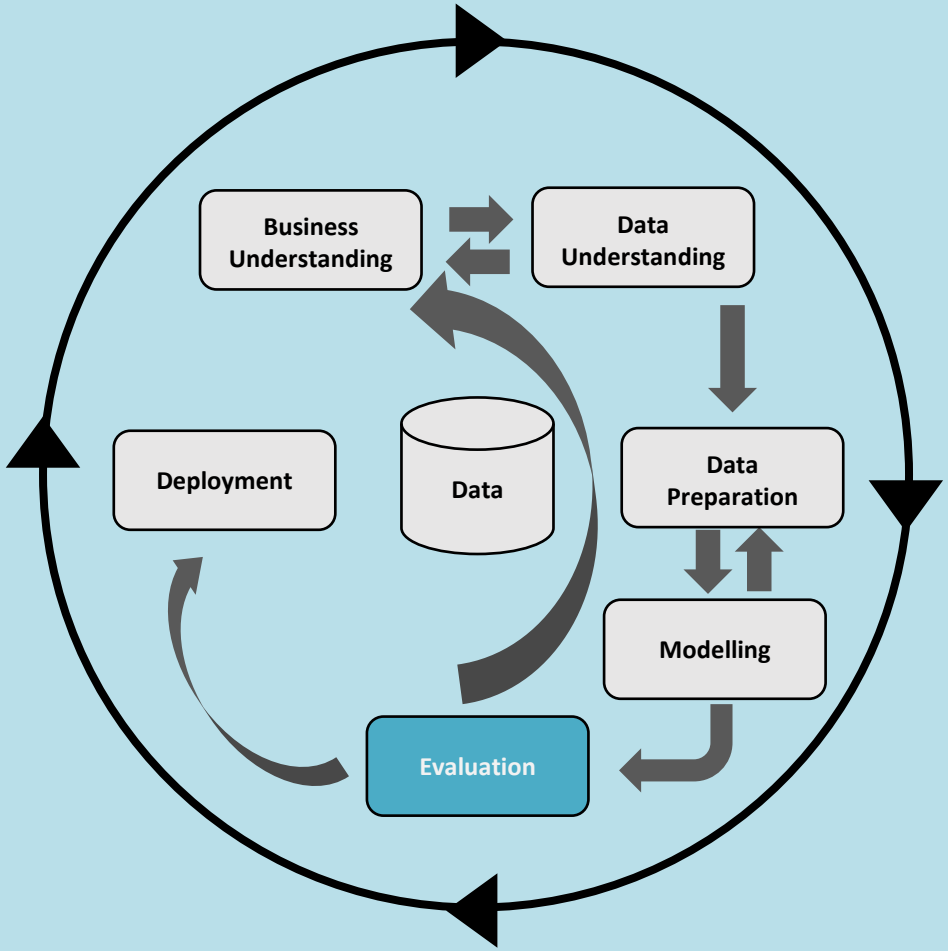


Chapman Pete et al. (1999)

11.1 CRISP-DM – Evaluation

- **What activities are related to the evaluation phase?**
  - Objectives achievement evaluation
  - Build measurement model

- **Other common activities:**
  - Make a presentation for management
  - Define measures
  - Compute business impact

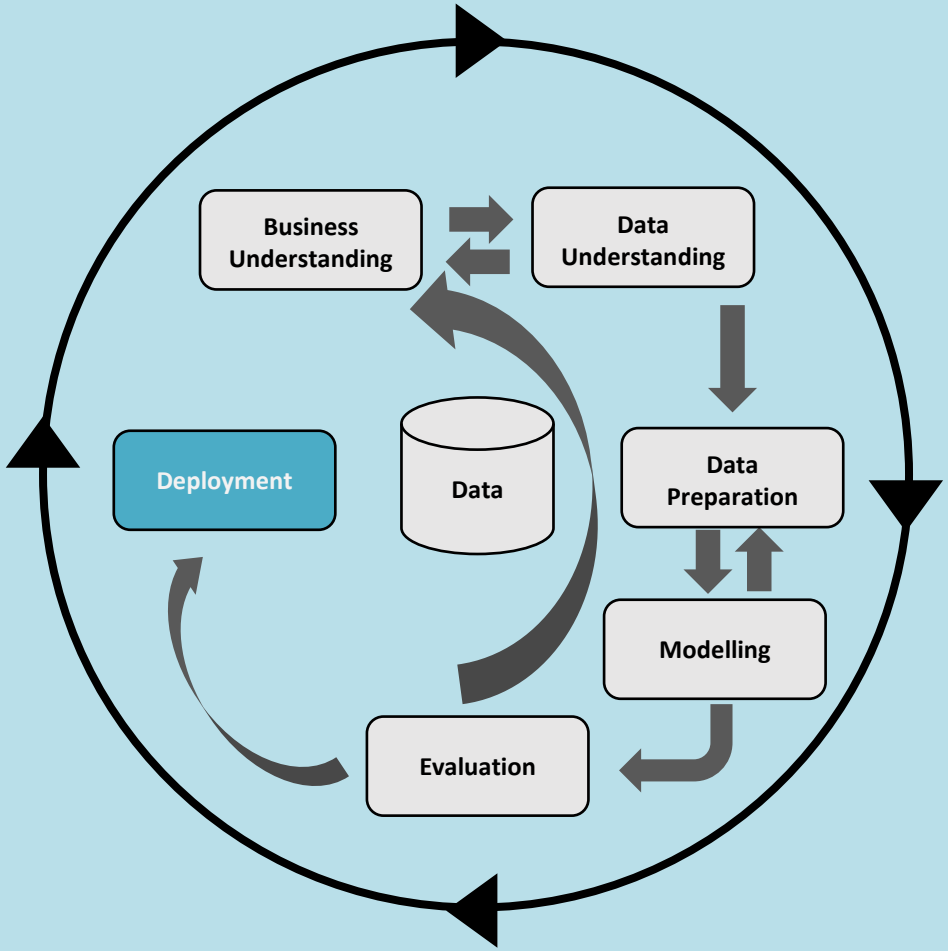


Chapman Pete et al. (1999)

11.1 CRISP-DM – Deployment

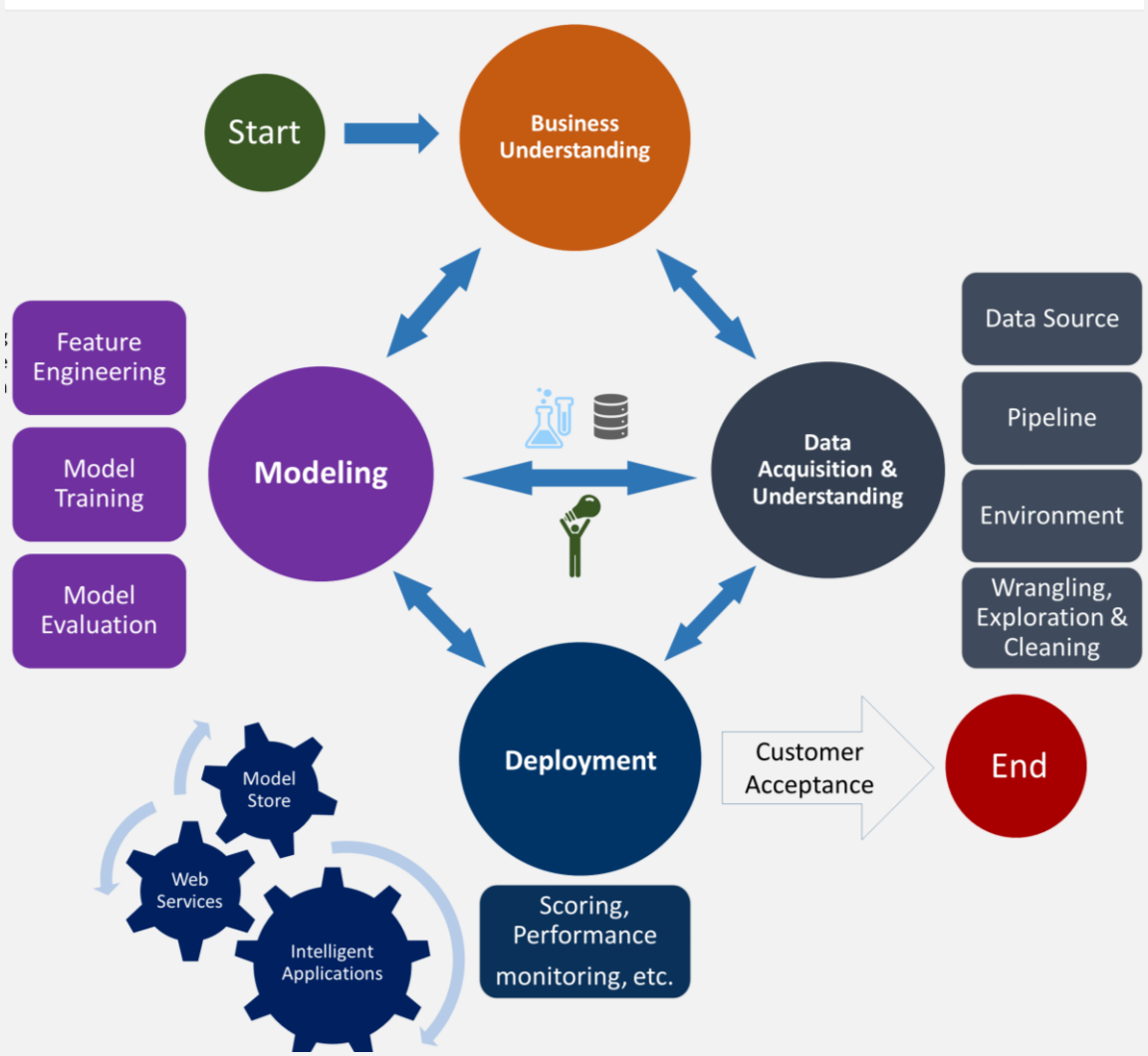
- **What activities are related to the deployment phase?**

  - Implement an AI-based Information System (result of model deployment)
  - Repeatable analytic process implementation
  - Communicate results



Chapman Pete et al. (1999)

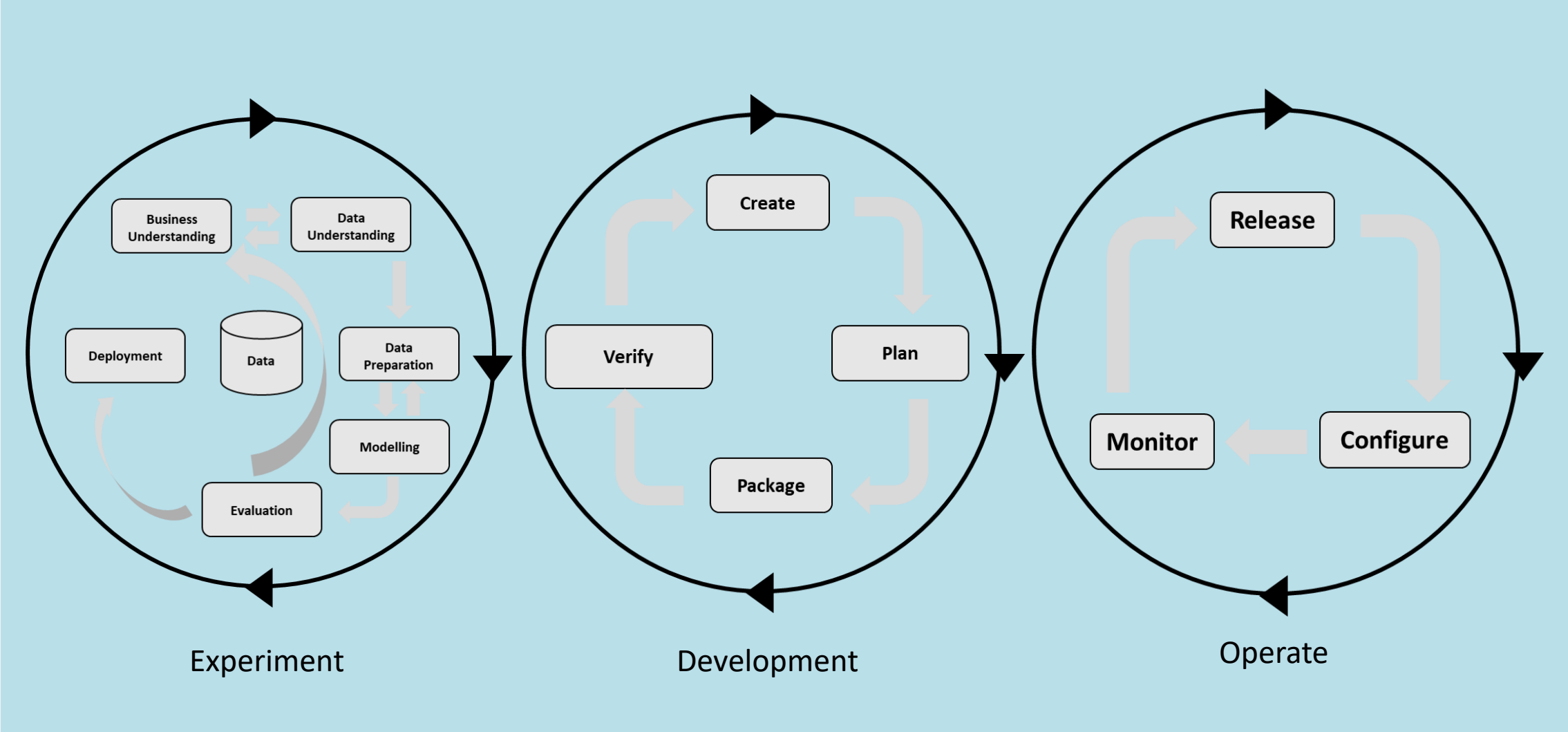
11.1 Microsoft Team Data Science Process (TDSP)



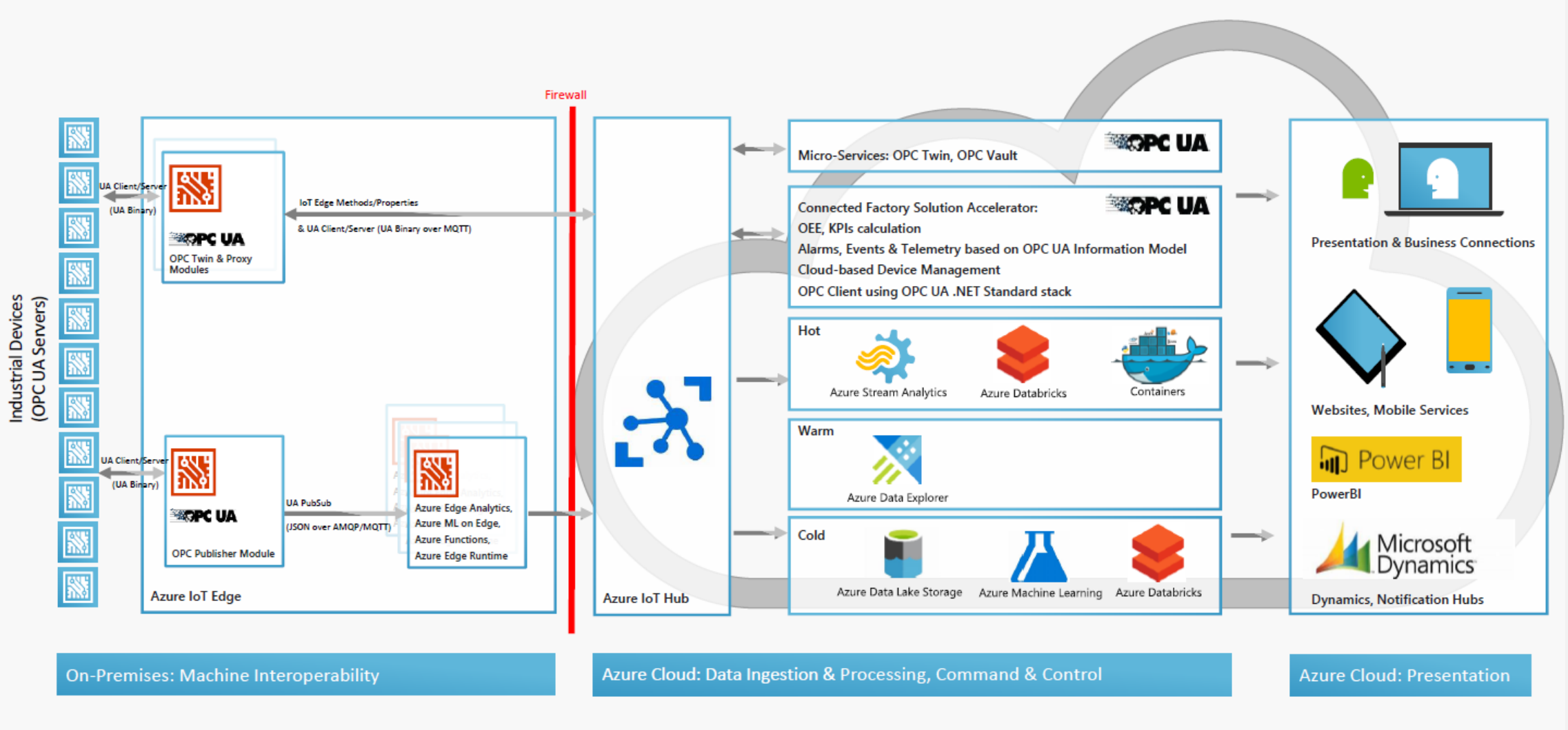
- **lifecycle that your team can use to structure your data science projects**
- **Stages respective to CRISP-DM**
  - Business Understanding: Define the project's objectives and success metrics in collaboration with stakeholders.
  - Data Acquisition and Understanding: Collect and explore relevant data to assess its quality and suitability for the project.
  - Modeling: Develop and train predictive models using the prepared data.
  - Deployment: Implement the models into a production environment for operational use.
  - Customer Acceptance: Validate the models with stakeholders to ensure they meet business requirements and deliver value.

Adapted from Microsoft (2020), online available https://learn.microsoft.com/en-us/azure/architecture/data-science-process/lifecycle

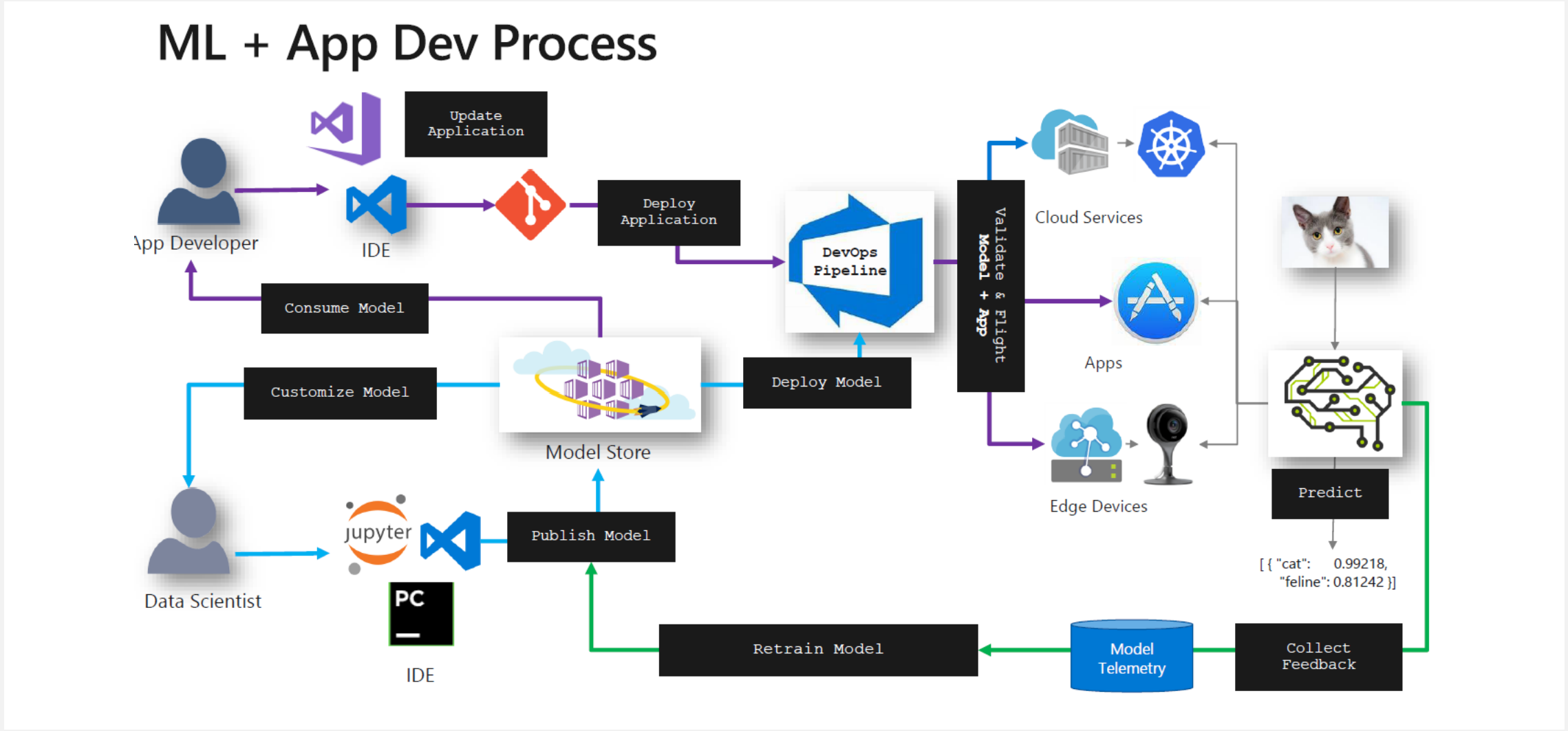
11.1 Lifecycle of AI-based Information Systems



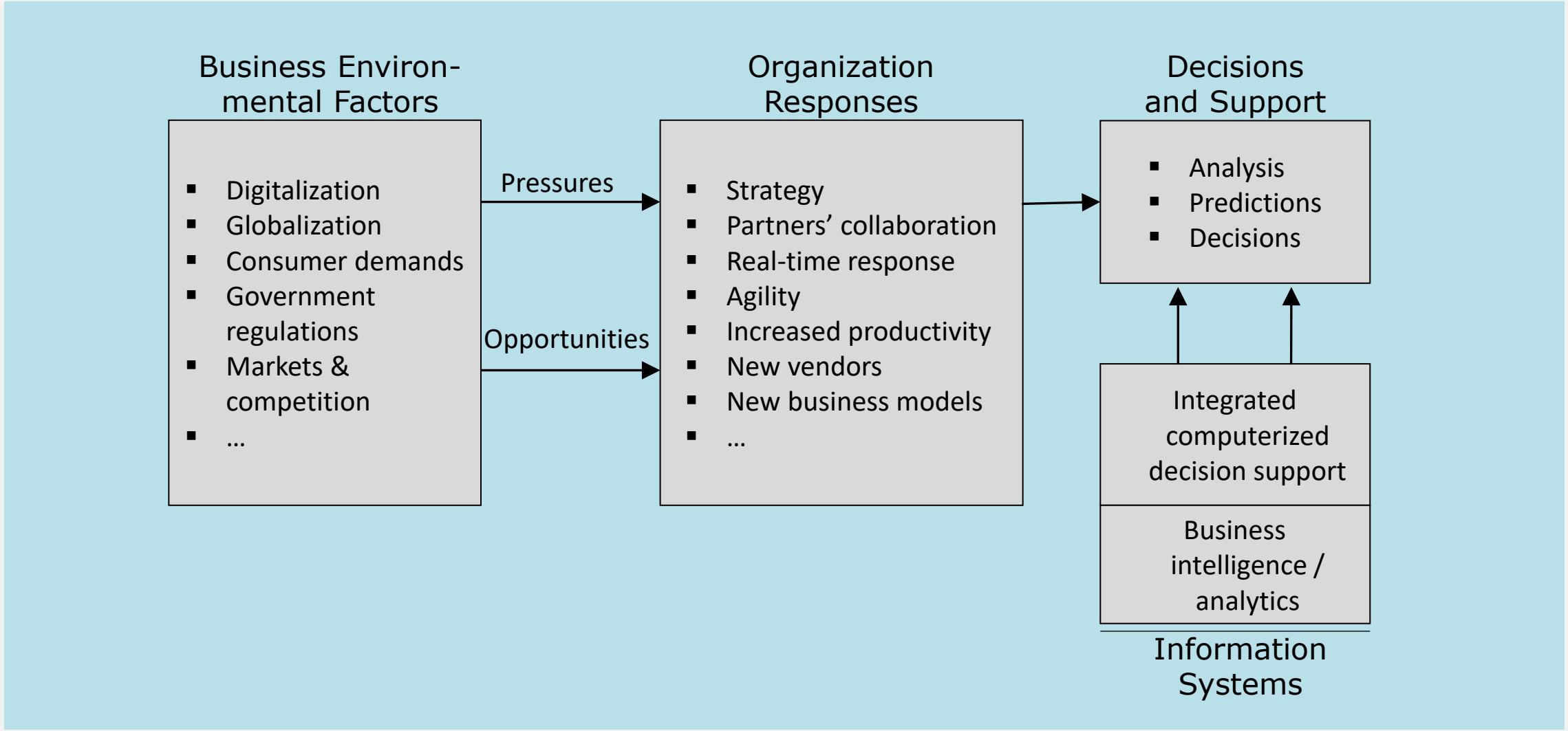
11.1 Integrated AI-Plattforms: Azure



11.1 Example: Azure ML Development Process



11.1 The Business Pressure-Response-Support Model



Based on Based on Sharda et al. (2014)

11.1 Win by Deployment



Rules of Machine Learning:

Best Practices for ML Engineering

Martin Zinkevich

- Deploy simple models in the spirit of winning by shipping
- Make quick iterations through the outermost development loop

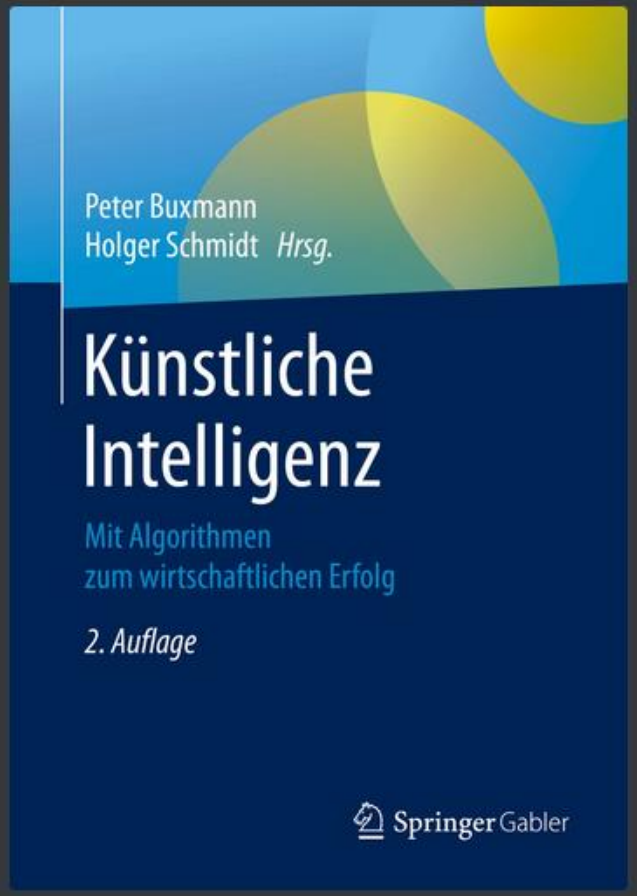
Based on Bernardi, L et al. (2019); Google (2019); Zinkevich, M (2019)

11.1 Deploy Models on Real Data as Fast as Possible

- Deploy a baseline model on production data as soon as possible
- Deploying your model on production data can be enlightening
- As a countermeasure, it's often a good idea to deploy a simple model on production data as soon as possible
- Deliver value incrementally and quickly
- Measure time to results, not results. Sometimes it can be tricky to get into the mindset of delivering value quickly.

Based on Bernardi, L et al. (2019); Google (2019); Zinkevich, M (2019)

11.1 Successful AI Projects: Book Recommendation

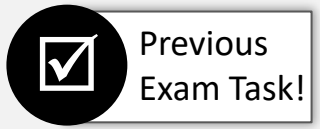


Künstliche Intelligenz wird als Basistechnologie ähnlich wie Elektrizität viele Unternehmen und Arbeitsplätze grundlegend verändern. Nach einer allgemein verständlichen Einführung in die technischen Grundlagen und ökonomischen Effekte zeigen zehn Unternehmen in der 2. Auflage unseres Buches, wie sie Algorithmen einsetzen, um Routinejobs zu automatisieren, aus Daten Erkenntnisse abzuleiten und neue Geschäftsmodelle zu entwickeln. Mit dem Buch wollen wir KI entmystifizieren, für Unternehmen handhabbar und für Interessierte verständlich machen. Wir wollen Entscheidern:innen Mut machen, sich mit dieser essentiellen Technologie stärker zu beschäftigen, damit der Vorsprung der amerikanischen und chinesischen Wettbewerber nicht noch größer wird. Mit Beiträgen von:

- Commerzbank: Künstliche Intelligenz in Banken
- Volkswagen: KI-Innovationen über das autonome Fahren hinaus
- Otto: Data Science im Online-Handel
- SAP: Effiziente Prozesse mit Künstlicher Intelligenz
- Microsoft: Künstliche Intelligenz in der Cloud
- Serviceware: Künstliche Intelligenz im Enterprise Service Management
- Software AG: Intelligentes IOT: Erkenntnisse aus IOT-D[illegible]ch Machine Learning
- IBM: Intelligente KI-Lösungen
- Empolis: Mit KI immer die richtigen Entscheidungen treffen
- Heraeus: KI-gestützte visuelle Inspektion in der Elektronikir

Künstliche Intelligenz: Mit Algorithmen zum wirtschaftlichen Erfolg (↗ki-business.de)

11.1 Classroom Task



Your turn!

Task

Please discuss with your neighbors:

- What is the difference between data understanding and business understanding? Why is there an interaction between these two phases?

Task

Please name three problems, we discussed in lecture, an AI Specialist should have in mind when he/she designs AI-based information systems?

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► **Duration:**
- 135 min + FAQs

► **Relevant for Exam:**
- 11.1-11.4

11.3 Guidelines for Human-AI Interaction from Microsoft

- The Researcher propose 18 general applicable design guidelines for human-AI interaction.
- Guidelines are validated through multiple rounds of evaluation including a user study where this guideline is tested against other popular AI-infused products.


Guidelines for Human-AI Interaction

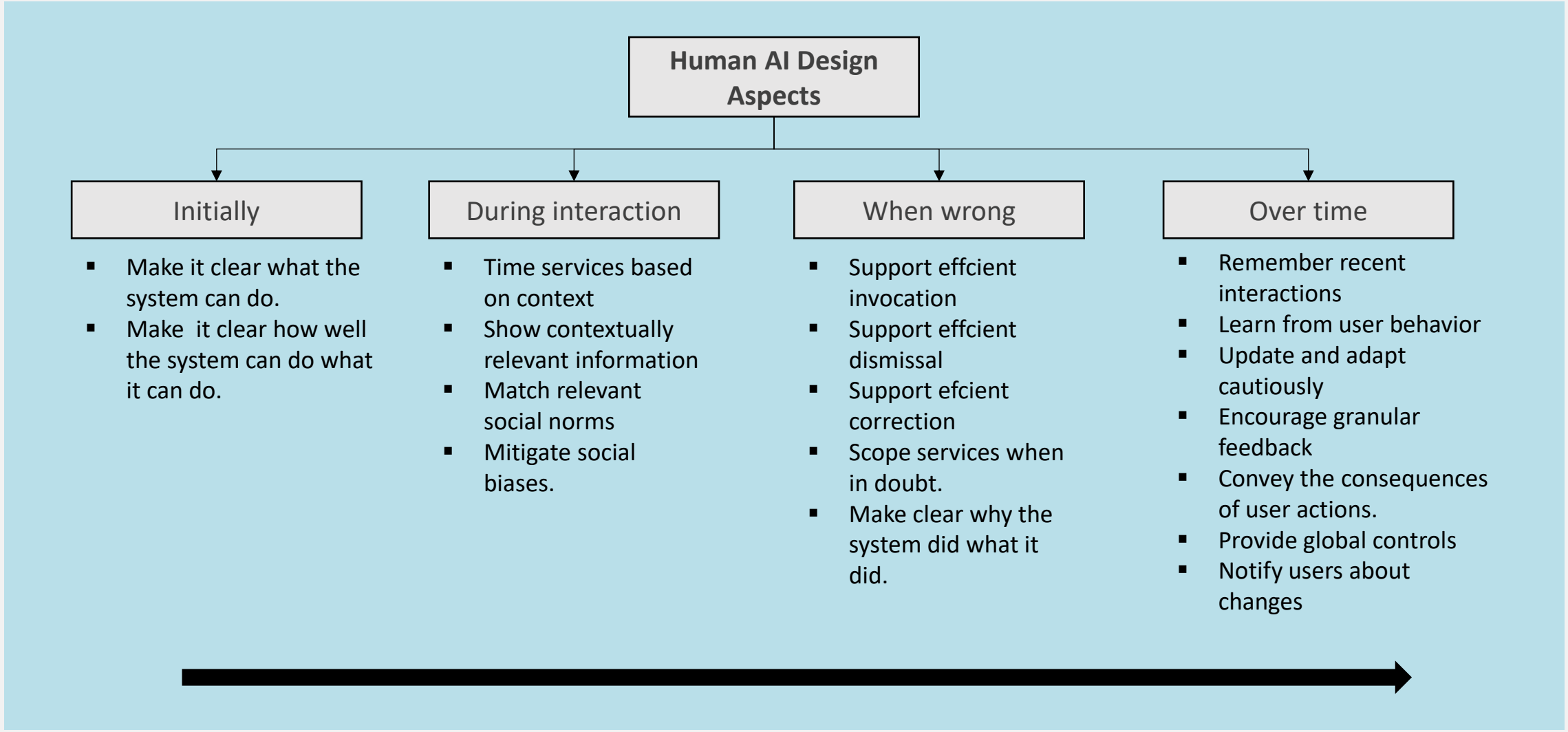
The following part of the lecture is mainly based on Microsoft's Best Practices for AI-Design ↗www.microsoft.com.

		AI Design Guidelines	Example Applications of Guidelines
Initially	G1	**Make clear what the system can do.** Help the user understand what the AI system is capable of doing.	[Activity Trackers, Product #1] "Displays all the metrics that it tracks and explains how. Metrics include movement metrics such as steps, distance traveled, length of time exercised, and all-day calorie burn, for a day."
	G2	**Make clear how well the system can do what it can do.** Help the user understand how often the AI system may make mistakes.	[Music Recommenders, Product #1] "A little bit of hedging language: 'we think you'll like'."
During interaction	G3	**Time services based on context.** Time when to act or interrupt based on the user's current task and environment.	[Navigation, Product #1] "In my experience using the app, it seems to provide timely route guidance. Because the map updates regularly with your actual location, the guidance is timely."
	G4	**Show contextually relevant information.** Display information relevant to the user's current task and environment.	[Web Search, Product #2] "Searching a movie title returns show times in near my location for today's date"
	G5	**Match relevant social norms.** Ensure the experience is delivered in a way that users would expect, given their social and cultural context.	[Voice Assistants, Product #1] "[The assistant] uses a semi-formal voice to talk to you - spells out "okay" and asks further questions."
	G6	**Mitigate social biases.** Ensure the AI system's language and behaviors do not reinforce undesirable and unfair stereotypes and biases.	[Autocomplete, Product #2] "The autocomplete feature clearly suggests both genders [him, her] without any bias while suggesting the text to complete."
When wrong	G7	**Support efficient invocation.** Make it easy to invoke or request the AI system's services when needed.	[Voice Assistants, Product #1] "I can say [wake command] to initiate."
	G8	**Support efficient dismissal.** Make it easy to dismiss or ignore undesired AI system services.	[E-commerce, Product #2] "Feature is unobtrusive, below the fold, and easy to scroll past...Easy to ignore."
	G9	**Support efficient correction.** Make it easy to edit, refine, or recover when the AI system is wrong.	[Voice Assistants, Product #2] "Once my request for a reminder was processed I saw the ability to edit my reminder in the UI that was displayed. Small text underneath stated 'Tap to Edit' with a chevron indicating something would happen if I selected this text."
	G10	**Scope services when in doubt.** Engage in disambiguation or gracefully degrade the AI system's services when uncertain about a user's goals.	[Autocomplete, Product #1] "It usually provides 3-4 suggestions instead of directly auto completing it for you"
	G11	**Make clear why the system did what it did.** Enable the user to access an explanation of why the AI system behaved as it did.	[Navigation, Product #2] "The route chosen by the app was made based on the Fastest Route, which is shown in the subtext."
Over time	G12	**Remember recent interactions.** Maintain short term memory and allow the user to make efficient references to that memory.	[Web Search, Product #1] "[The search engine] remembers the context of certain queries, with certain phrasing, so that it can continue the thread of the search (e.g., 'who is he married to' after a search that surfaces Benjamin Bratt)"
	G13	**Learn from user behavior.** Personalize the user's experience by learning from their actions over time.	[Music Recommenders, Product #2] "I think this is applied because every action to add a song to the list triggers new recommendations."
	G14	**Update and adapt cautiously.** Limit disruptive changes when updating and adapting the AI system's behaviors.	[Music Recommenders, Product #2] "Once we select a song they update the immediate song list below but keeps the above one constant."
	G15	**Encourage granular feedback.** Enable the user to provide feedback indicating their preferences during regular interaction with the AI system.	[Email, Product #1] "The user can directly mark something as important, when the AI hadn't marked it as that previously."
	G16	**Convey the consequences of user actions.** Immediately update or convey how user actions will impact future behaviors of the AI system.	[Social Networks, Product #2] "[The product] communicates that hiding an Ad will adjust the relevance of future ads."
	G17	**Provide global controls.** Allow the user to globally customize what the AI system monitors and how it behaves.	[Photo Organizers, Product #1] "[The product] allows users to turn on your location history so the AI can group photos by where you have been."
	G18	**Notify users about changes.** Inform the user when the AI system adds or updates its capabilities.	[Navigation, Product #2] "[The product] does provide small in-app teaching callouts for important new features. New features that require my explicit attention are pop-ups."

Table 1: Our 18 human-AI interaction design guidelines, roughly categorized by when they likely are to be applied during interaction with users, along with illustrative applications (rated as "clearly applied" by participants) across products tested by participants in our user study.

Adapted from Amershi, S et al. (2019)

11.3 AI Design Guidelines



Adapted from Amershi, S et al. (2019)

11.3 AI Design Guideline: Initially

1

G1 - Make clear what the system can do.

- Help the user understand what the AI system is capable of doing
- **Example:** [Activity Trackers, Product #1] “Displays all the metrics that it tracks and explains how. Metrics include movement metrics such as steps, distance traveled, length of time exercised, and all-day calorie burn, for a day.”
- **Example:** [Photo Organizers, Product #1] “We know the AI is able to detect and associate an image with a category, but the user does not know all the categories available.”

Example: User Onboarding

Here you see <functionality>!

Try out to…

This button allows you to <function>

Adapted from Amershi, S et al. (2019)

11.3 AI Design Guideline: Initially

2

G2 - Make clear how well the system can do what it can do.

- Help the user understand how often the AI system may make mistakes.
- **Example:** [Music Recommenders, Product #1] “A little bit of hedging language: ‘we think you’ll like’.”
- **Example:** [Voice Assistants, Product #1] “Aside from the ‘Hi, how can I help?’, [the product] does not promise anything more. No expectation of quality is set.”

Example: Message / show how often you used the functionality of the system

9 of 10
Warnings
correct

Number of correct cases

Adapted from Amershi, S et al. (2019)

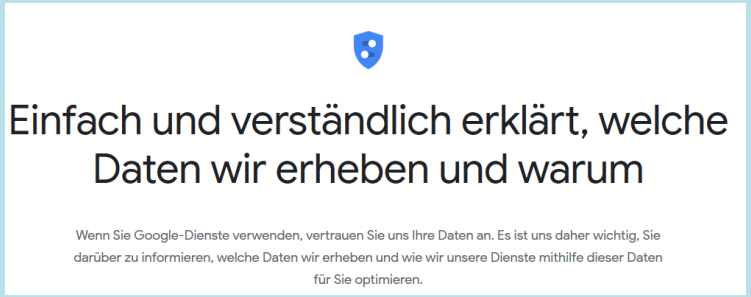
11.3 AI Design Guideline: During interaction

3

G3 - Time services based on context.

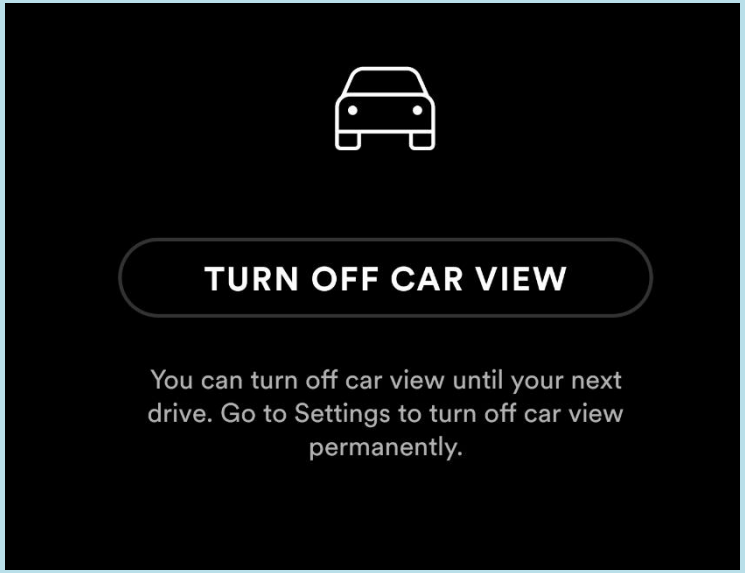
- Time when to act or interrupt based on the user's current task and environment
- **Example:** [Autocomplete, Product #1] "Suggestions are always present when you might need them (whenever the keyboard is up)"
- **Example:** [Social Networks, Product #1] "If the user has not accessed [the product] in a while, the application will let the user know that there is something new to be explored - a story, video, etc."

Example: User-sensitive systems









Adapted from Amershi, S et al. (2019)

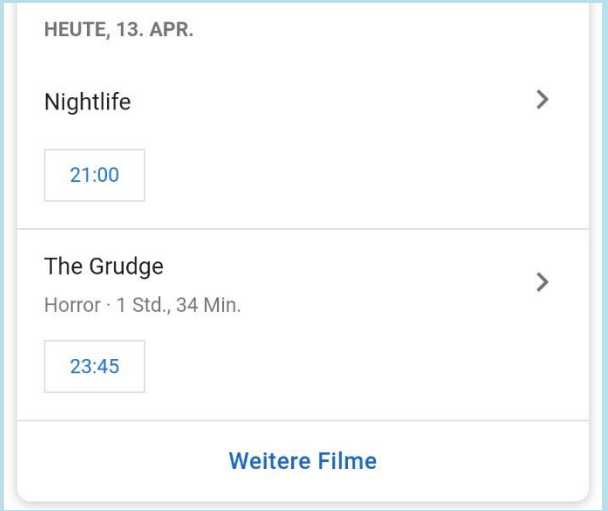
11.3 AI Design Guideline: During interaction

4

G4 - Show contextually relevant information.

- Display information relevant to the user's current task and environment.
- **Example:** [Web Search, Product #2] "Searching a movie title returns show times near my location for today's date"
- **Example:** [Navigation, Product #1] "When I use [the product] for driving directions, it remembers where I parked my car. Next time when I open the app, it suggests routing me back to my car."

Example: Search a recent movie in the google search of your android phone and it returns playtimes of your local cinema.



Adapted from Amershi, S et al. (2019)

11.3 AI Design Guideline: During interaction

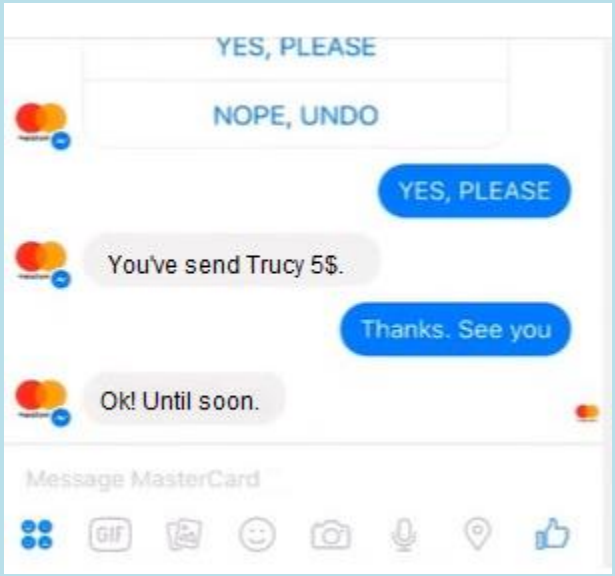
5

G5 - Match relevant social norms

- Ensure the experience is delivered in a way that users would expect, given their social and cultural context.
- **Example:** [Voice Assistants, Product #1] "[The assistant] uses a semiformal voice to talk to you - spells out "okay" and asks further questions."
- **Example:** [Navigation, Product #1] "If you select walking, the AI avoid(s) busy roads and searches for trails."

Example: Example Mastercard chatbot interface

(I know like most chatbots this is no 100% AI example)



Adapted from Amershi, S et al. (2019)

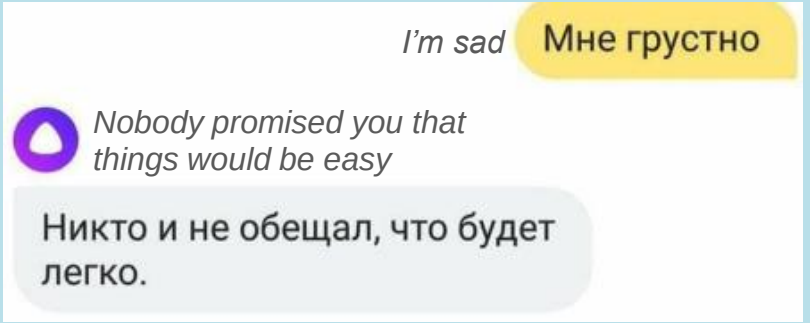
11.3 AI Design Guideline: During interaction

6

G6 - Mitigate social biases.

- Ensure the AI system's language and behaviors do not reinforce undesirable and unfair stereotypes and biases.
- **Example:** [Web Search, Product #2] "a search for CEO or Doctor shows somewhat diverse people in the resulting images...The images are pretty diverse in terms of gender and ethnicity, although still lack in some respects such as disability"
- **Example:** [Autocomplete, Product #2] "The autocomplete feature clearly suggests both genders [him, her] without any bias while suggesting the text to complete."

Example: Yandex Assistant



Adapted from Amershi, S et al. (2019)

11.3 AI Design Guideline: When wrong

7

G7 - Support efficient invocation.

- Make it easy to invoke or request the AI system's services when needed.
- **Example:** [Voice Assistants, Product #1] "I can say [wake command] to initiate."
- **Example:** [E-commerce, Product #1] "In addition to the system giving you recommendations as you browse, you can go to your "Browsing history > Manage history > More like this" to get recommendations specific to a particular product."

Example: Google Work

If you add your workplace and your home, Google warns you if there are any unexpected events (e.g. traffic jam)

Furthermore, it adds your standard ways to the app interface





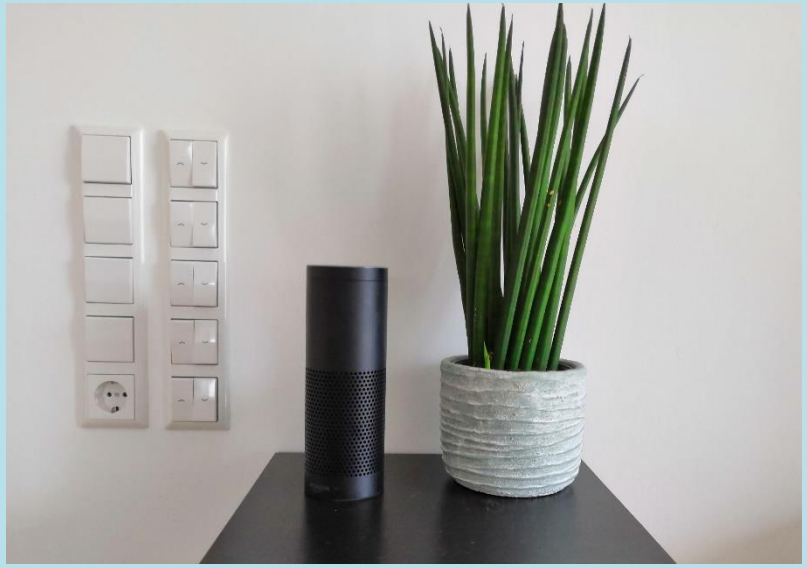
Adapted from Amershi, S et al. (2019)

11.3 AI Design Guideline: When wrong

8

G8 - Support efficient dismissal

- Make it easy to dismiss or ignore undesired AI system services.
- **Example:** [E-commerce, Product #2] “Feature is unobtrusive, below the fold, and easy to scroll past...Easy to ignore.”
- **Example:** [Voice Assistants, Product #1] “I can say “nevermind” to dismiss it once I have said [wake command]. I can also just not say anything and it stops listening.”



Stop or pause:
- "Alexa, stop" or,
- "Alexa, shut up."

Mute or unmute:
- "Alexa, mute" or,
- "Alexa, unmute."

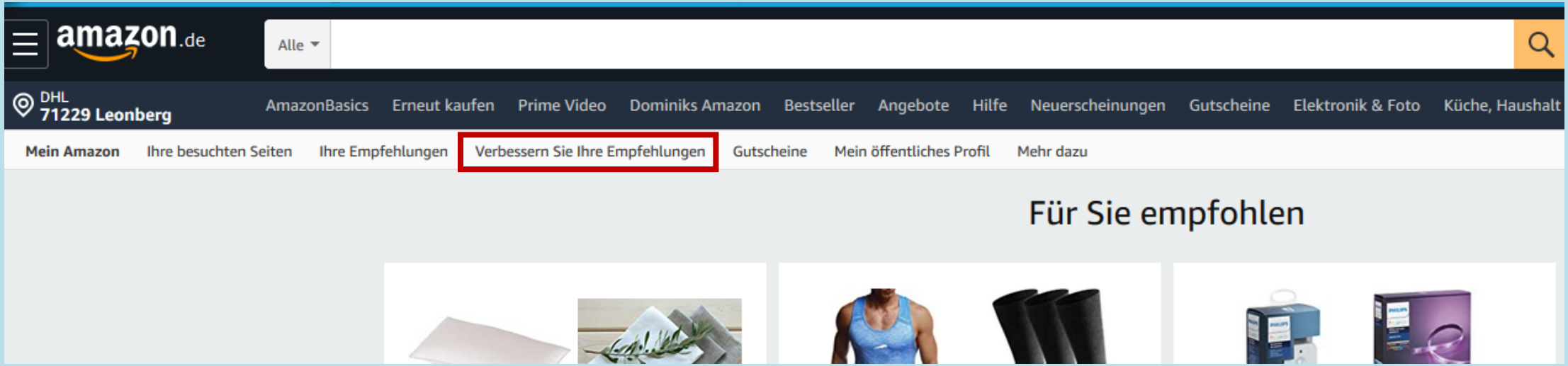
Adapted from Amershi, S et al. (2019)

11.3 AI Design Guideline: When wrong

9

G9 - Support efficient correction.

- Make it easy to edit, refine, or recover when the AI system is wrong.
- **Example:** [Navigation, Product #1] “If [the product] is wrong about where I parked my car, it provides an easy way to edit the location by dragging on the map.”
- **Example:** [Web Search, Product #2] “automatically ‘corrects’ spelling errors, etc. but gives option at top to return to query as originally typed...Notes that the query had been corrected and is one click to revert back to original”



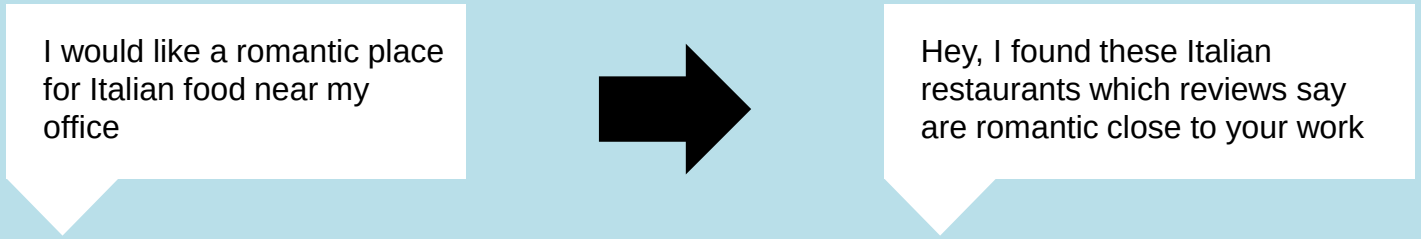
Adapted from Amershi, S et al. (2019)

11.3 AI Design Guideline: When wrong

10

G10 - Scope services when in doubt.

- Engage in disambiguation or gracefully degrade the AI system's services when uncertain about a user's goals.
- **Example:** [Navigation, Product #1] "If more than one line takes the same route the user can choose between the preferred line."
- **Example:** [Autocomplete, Product #1] "It usually provides 3-4 suggestions instead of directly auto completing it for you"



Adapted from Amershi, S et al. (2019)

11.3 AI Design Guideline: When wrong

11

G11 - Make clear why the system did what it did.

- Enable the user to access an explanation of why the AI system behaved as it did.
- **Example:** [E-commerce, Product #1] “Clicking “Why recommended” explains why they have recommended that particular item to you.”
- **Example:** [Music Recommenders, Product #2] “I think this applies because each of recommendation has some information as to which songs are displayed on it - similar to the song, from the same artist, from the same album etc.”





Adapted from Amershi, S et al. (2019)

11.3 AI Design Guideline: Over time

12

G12 - Remember recent interactions.

- Maintain short term memory and allow the user to make efficient references to that memory.
- **Example:** [Navigation, Product #1] “Opening the app shows a list of recent destinations, as well as allows you to access “favorite” locations.”
- **Example:** [Voice Assistants, Product #1] “[The assistant] seems to remember conversation context at least one command back. When asked “[wake command], what’s the reminder?” she announces the last unheard remind me.”

Resume the last played audiobook: "Alexa, resume my book."



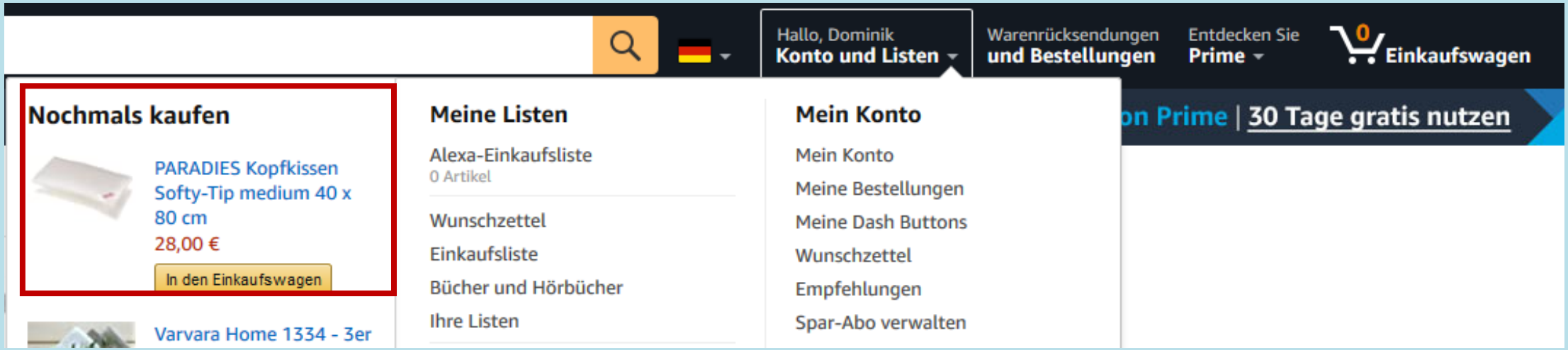
Adapted from Amershi, S et al. (2019)

11.3 AI Design Guideline: Over time

13

G13 - Learn from user behavior.

- Personalize the user's experience by learning from their actions over time.
- **Example:** [Music Recommenders, Product #2] "I think this is applied because every action to add a song to the list triggers new recommendations."
- **Example:** [Email, Product #1] "(My guess is) the system learns from what previous emails have attracted more attention from me (i.e. longer/more frequent reply, reading time taken, longer email threads, etc.) and infer email importance."



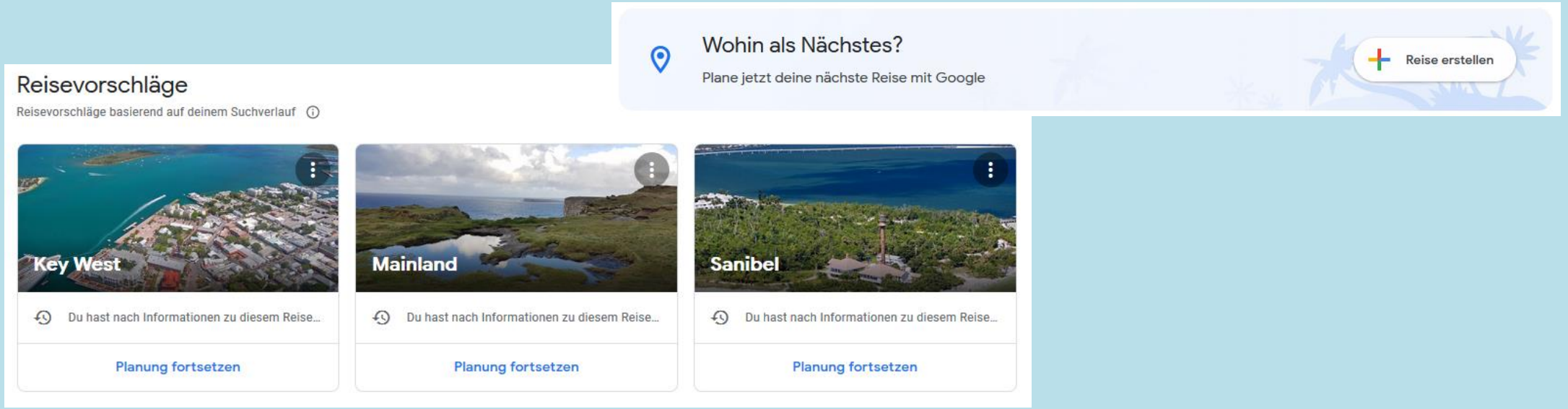
Adapted from Amershi, S et al. (2019)

11.3 AI Design Guideline: Over time

14

G14 - Update and adapt cautiously

- Limit disruptive changes when updating and adapting the AI system's behaviors.
- **Example:** [Music Recommenders, Product #2] "Once we select a song they update the immediate song list below but keeps the above one constant."
- **Example:** [Social Networks, Product #1] "Think this is good. When I unfollow someone it shows there stuff for a little bit? But after a day or so its gone. Or once I reload. But not RIGHT away."



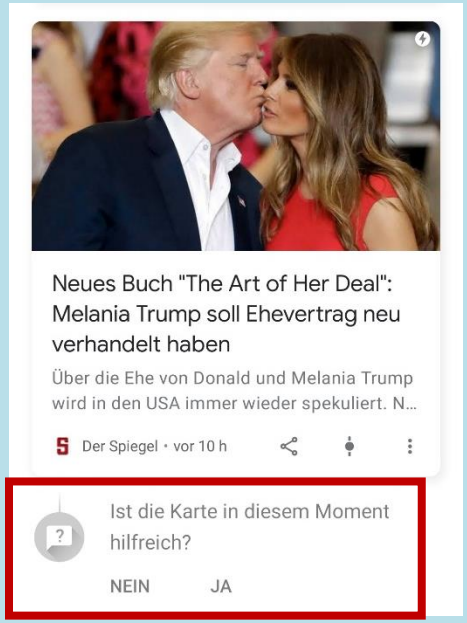
Adapted from Amershi, S et al. (2019)

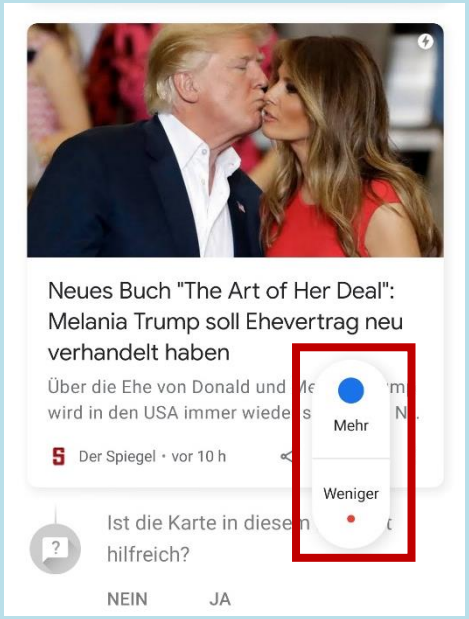
11.3 AI Design Guideline: Over time

15

G15 - Encourage granular feedback.

- Enable the user to provide feedback indicating their preferences during regular interaction with the AI system.
- **Example:** [Music Recommenders, Product #1] “Love/dislike buttons are prominent and easily accessible.”
- **Example:** [Email, Product #1] “The user can directly mark something as important, when the AI hadn’t marked it as that previously.”





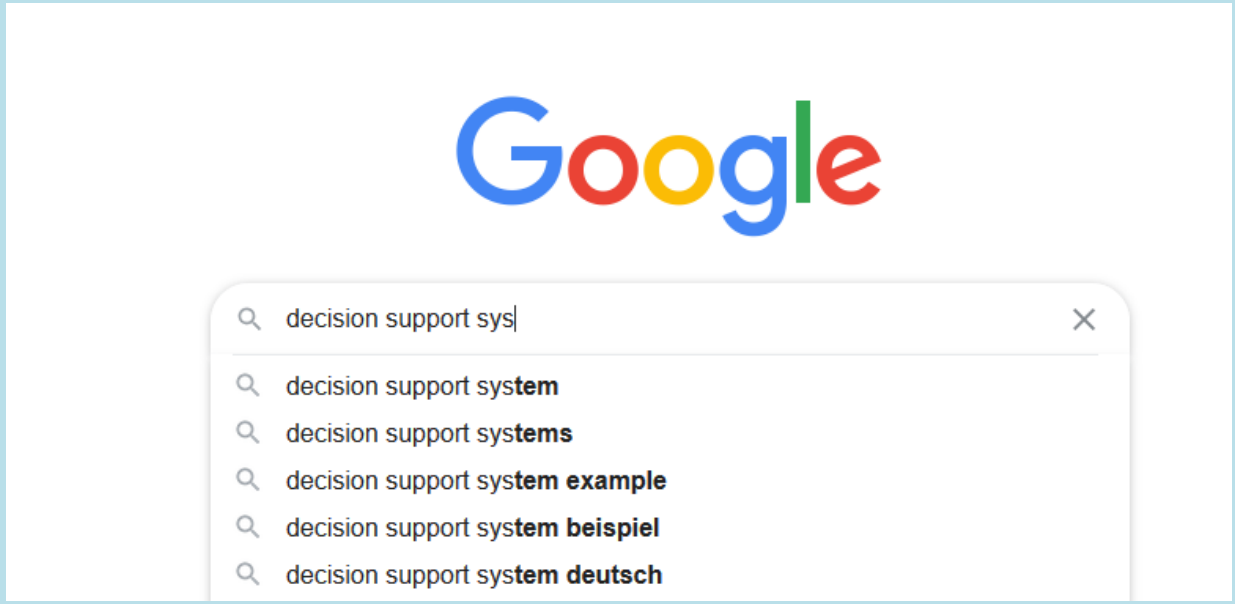
Adapted from Amershi, S et al. (2019)

11.3 AI Design Guideline: Over time

16

G16 - Convey the consequences of user actions.

- Immediately update or convey how user actions will impact future behaviors of the AI system.
- **Example:** [Music Recommenders, Product #1] “Tapping the like/dislike button results in immediate popups informing that the user will receive more/fewer recommendations like it.”
- **Example:** [Web Search, Product #1] “With different filters, the search results are auto updated.”



Adapted from Amershi, S et al. (2019)

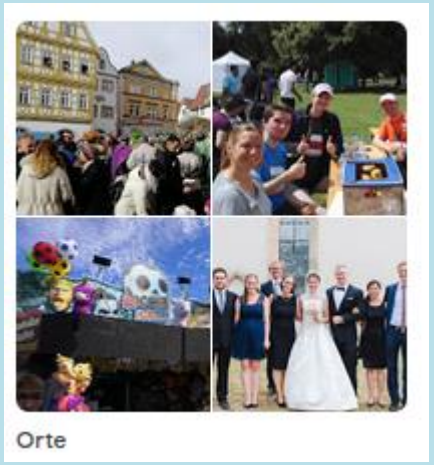
11.3 AI Design Guideline: Over time

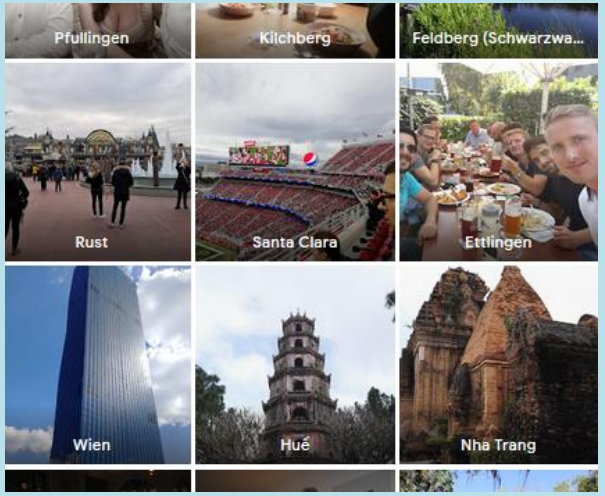
17

G17 - Provide global controls

- Allow the user to globally customize what the AI system monitors and how it behaves.
- **Example:** [Web Search. Product #2] “It has settings such as...private results that help users get results that are more relevant to them.”
- **Example:** [Photo Organizers, Product #1] “[The product] allows users to turn on your location history so the AI can group photos by where you have been.”







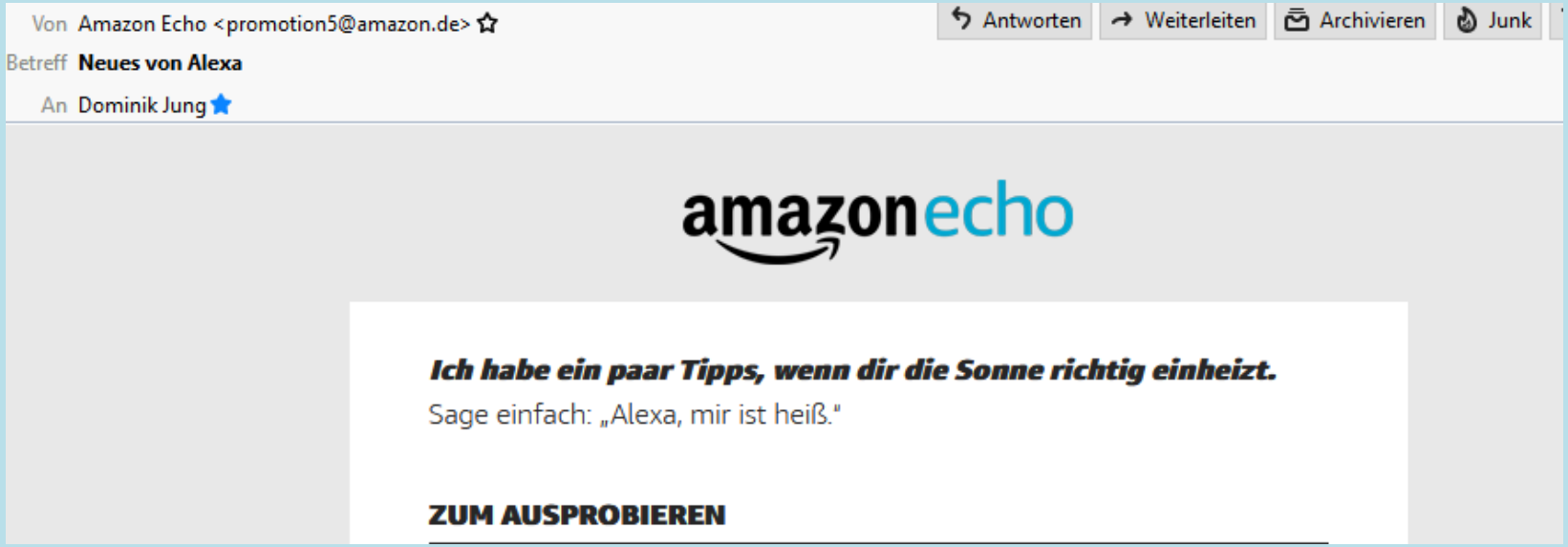
Adapted from Amershi, S et al. (2019)

11.3 AI Design Guideline: Over time

18

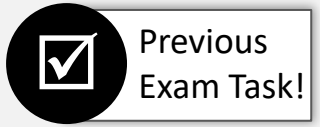
G18 - Notify users about changes.

- Inform the user when the AI system adds or updates its capabilities.
- **Example:** [Email, Product #2] “The help tab for the interface features a “What’s new” section which could be used to inform the user about AI system additions or capability updates.”
- **Example:** [Navigation, Product #2] “I don’t have a way to show this, but it does provide small in-app teaching callouts for important new features. New features that require my explicit attention are pop-ups.”



Adapted from Amershi, S et al. (2019)

11.3 Classroom Task



Your turn!

Task

Please name examples from real-life AI applications for the following design recommendations:

- G3 - Time services based on context.
- G7 - Support efficient invocation.
- G14 - Update and adapt cautiously

Outline

11 Building Productive AI-based Systems

- 11.1 The Challenge of Applied AI
- 11.2 Robotic Agents
- 11.3 Human-AI Interaction
- 11.4 AI Ethics and Governance
- 11.5 How to Continue your AI Journey

Exam Preparation and Course Repetitorium

Capstone Project / Case Challenge

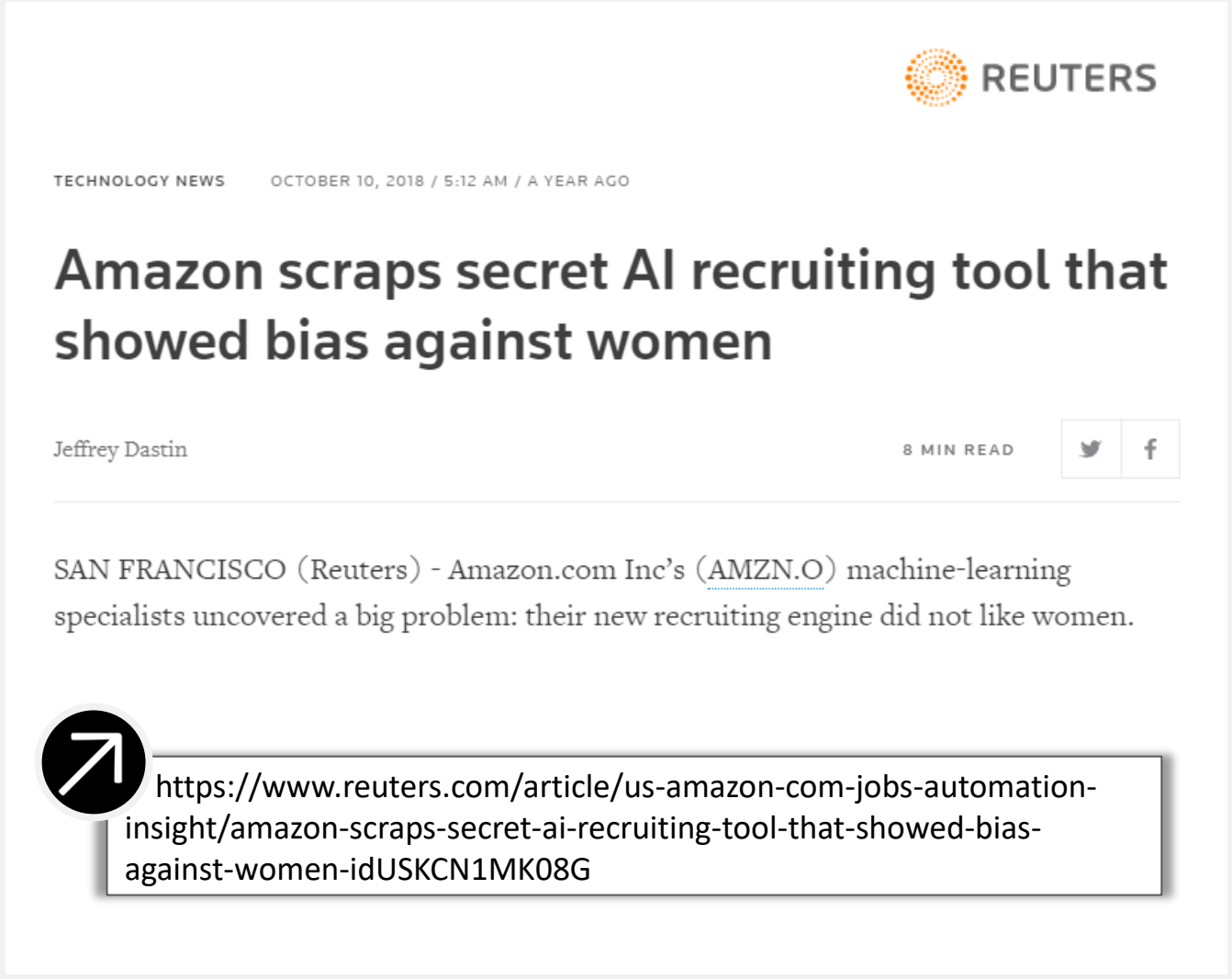


► **What you will learn:**
- Why many AI-based Information systems and intelligent agents fail in practice and how to tackle common problems in implementing AI-based information systems and intelligent agents
- Outlook and Future steps if you are interested in an AI-job

► **Duration:**
- 135 min + FAQs

► **Relevant for Exam:**
- 11.1-11.4

11.4 Amazon AI Recuriter doesn‘t like Women



REUTERS

TECHNOLOGY NEWS OCTOBER 10, 2018 / 5:12 AM / A YEAR AGO

Amazon scraps secret AI recruiting tool that showed bias against women

Jeffrey Dastin 8 MIN READ

SAN FRANCISCO (Reuters) - Amazon.com Inc's (AMZN.O) machine-learning specialists uncovered a big problem: their new recruiting engine did not like women.

https://www.reuters.com/article/us-amazon-com-jobs-automation-insight/amazon-scraps-secret-ai-recruiting-tool-that-showed-bias-against-women-idUSKCN1MK08G

- Amazons “AI Recruting Information System” had to be scrapped after showing a distinct bias against woman
- Amazon’s system taught itself that male candidates were preferable.
- It penalized resumes that included the word “women’s,” as in “women’s chess club captain.”
- It downgraded graduates of two all-women’s colleges, according to people familiar with the matter.

11.4 Biased AI Models Have Real-World Consequences

YT Fact Check Algorithm Fails



COMPAS



Google Photo Fail 2015



- Bias, as in “our model differs systematically between the expected prediction and the true value”, is a machine learning problem
- Bias, as in “a model reflecting undue prejudice in its predictions” is **not** simply a machine learning problem

Image source: COMPAS ↗Pixabay (2019) / ↗CC0

11.4 Biased Data is a Huge Challenge in AI Design



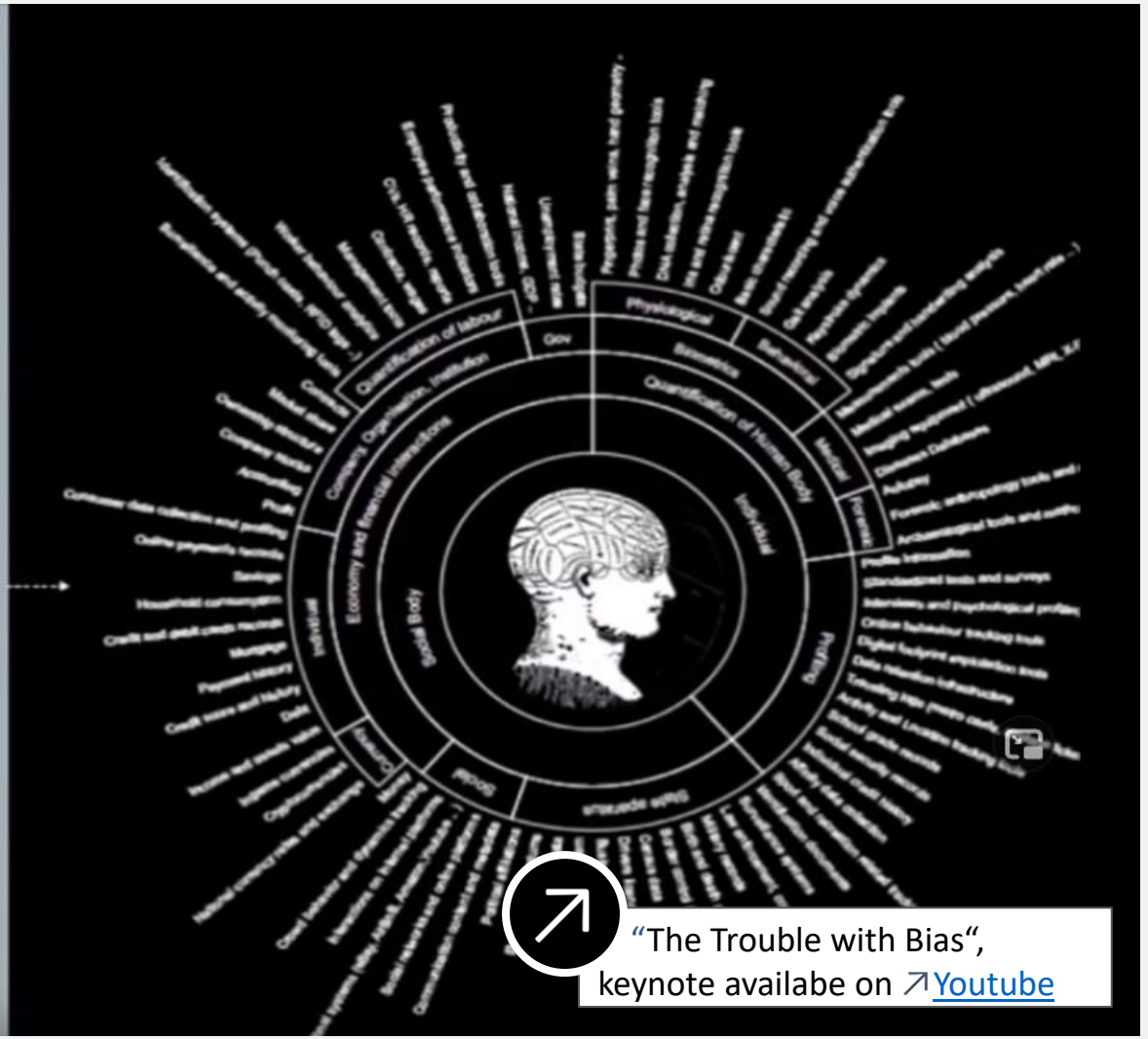
Image source: ↗Associate Professor Kate Crawford, University of New South Wales (2009) by andresmh from Wikimedia / ↗CC BY-SA 2.0

11.4 The Trouble with Bias - NIPS 2017 Keynote - Kate Crawford

THE TROUBLE WITH BIAS

KATE CRAWFORD

Distinguished Research Professor, NYU
Principal Researcher, Microsoft Research
Co-founder, AI Now Institute
@katecrawford



"The Trouble with Bias",
keynote availabe on ↗Youtube

11.4 Harm that AI Can Cause

- A “harm” is caused when a prediction or end outcome negatively impacts
  - an individual’s ability to establish their rightful personhood (harms of representation),
  - or leading to or independently impacting their ability to access resources (harms of allocation)

- **Implication**: Incorrectly representing individual’s in AI-based information systems, or any consecutive decision made by these systems in regards to individual

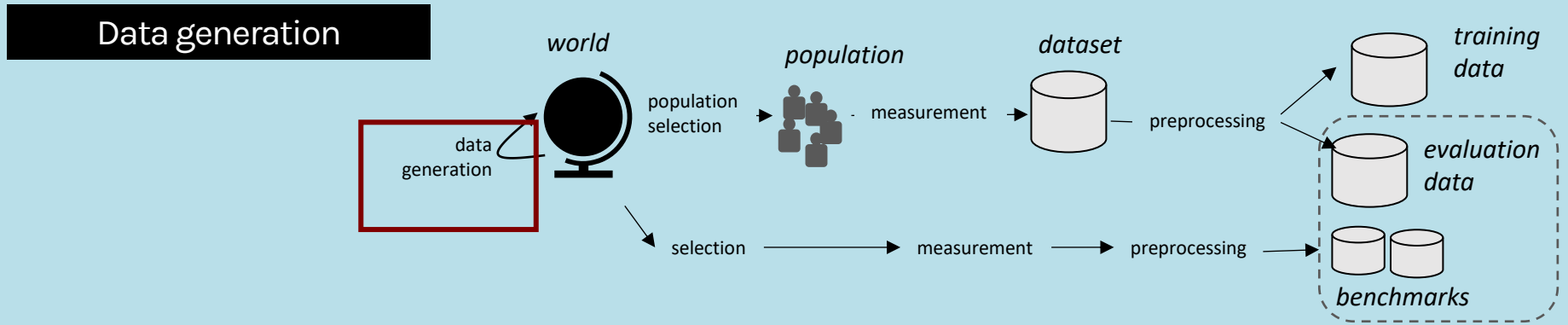
11.4 Ethics in Artificial Intelligence

- **Ethics**: Dealing with right vs. wrong, and moral obligations and duties of humans
- How right, how fair and how just, is the output, outcome and impact?
- Being answerable to these constitute moral obligations and duties of AI developers

11.4 Ethical Issues in Artificial Intelligence

- **What AI is**
  - Bias and fairness
  - Accountability
  - Transparency and understandability
  - Interpretability
  - Decision-maker/support
- **What AI does**
  - Work- and process security
  - Cyber-security and malicious use
  - Privacy
  - Human-AI interaction
- **What AI impacts**
  - Jobs and labor trends
  - Impact to modern society
  - Human-human interaction
  - Communication
  - Law
  - Politics
- **What AI can be**
  - Singularity?
  - Robot rights (see e.g Assimov)?
  - Part of society?

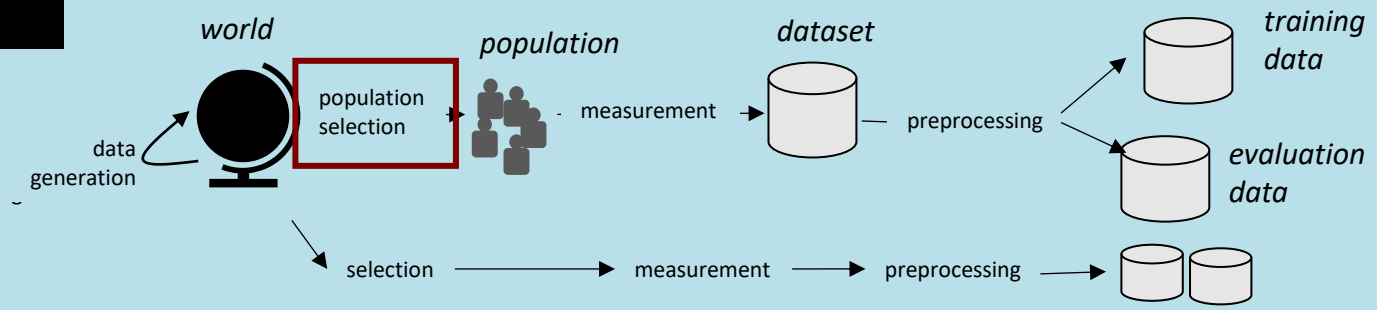
11.4 Historical Bias



- Historical bias arises when there is a misalignment between world as it is and the values or objectives to be encoded and propagated in a model.
- It is a normative concern with the state of the world, and exists even given perfect sampling and feature selection.

11.4 Representation Bias

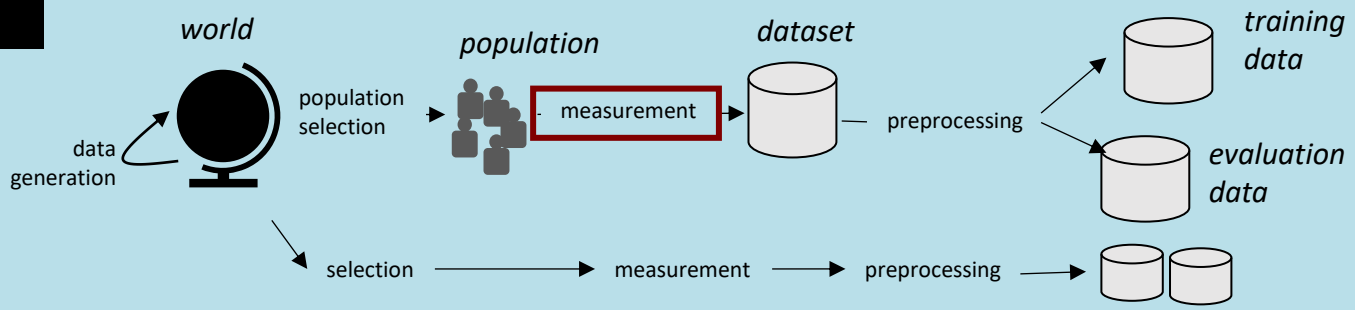
Data generation



- Representation bias arises while defining and sampling a development population
- It occurs when the development population under-represents, and subsequently causes worse performance, for some part of the final population

11.4 Measurement Bias

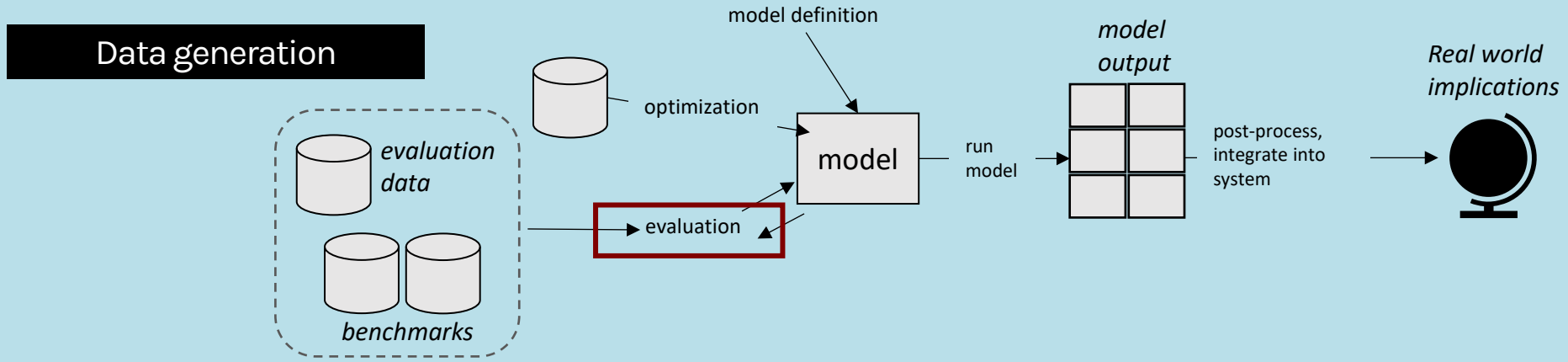
Data generation



Measurement bias arises when choosing and measuring the particular features and labels of interest

- Features considered to be relevant to the outcome are chosen, but these can be incomplete or contain group- or input dependent noise.
- In many cases, the choice of a single label to create a classification task may be an oversimplification that more accurately measures the true outcome of interest for certain groups.

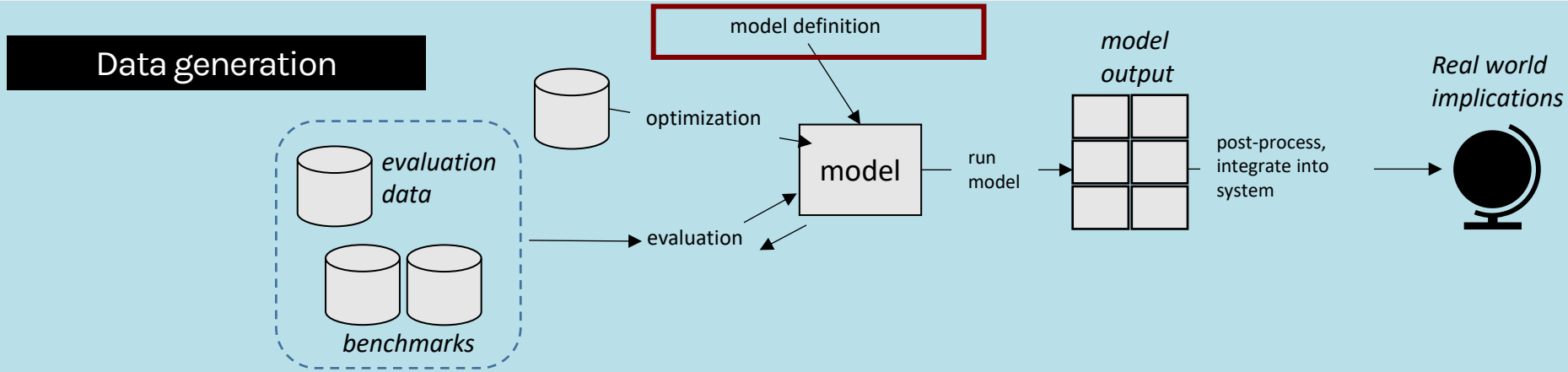
11.4 Evaluation Bias



Evaluation bias occurs during model iteration and evaluation, when the testing or external benchmark populations do not equally represent the various parts of the final population

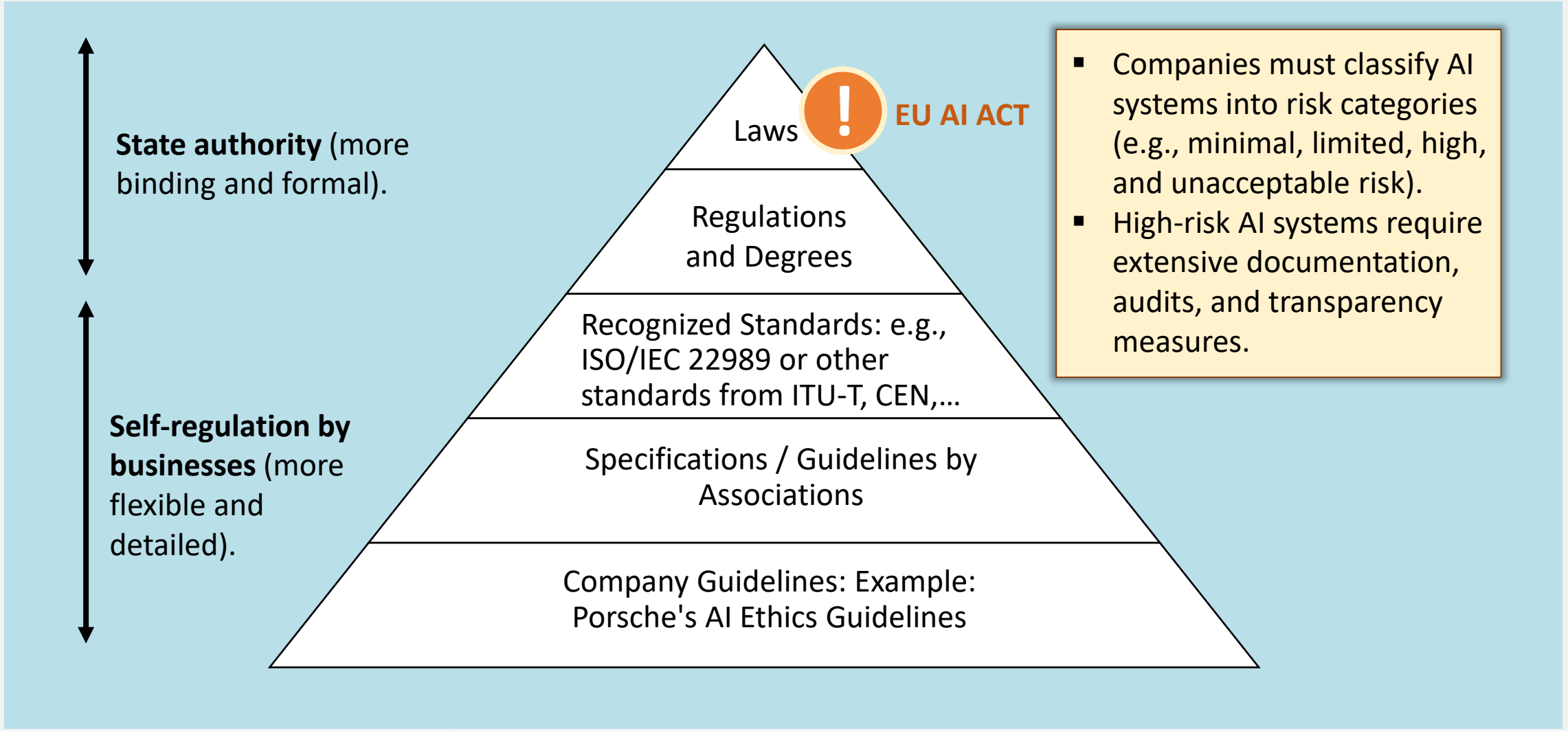
- Evaluation bias can also arise from the use of performance metrics that are not granular or comprehensive enough.

11.4 Aggregation Bias



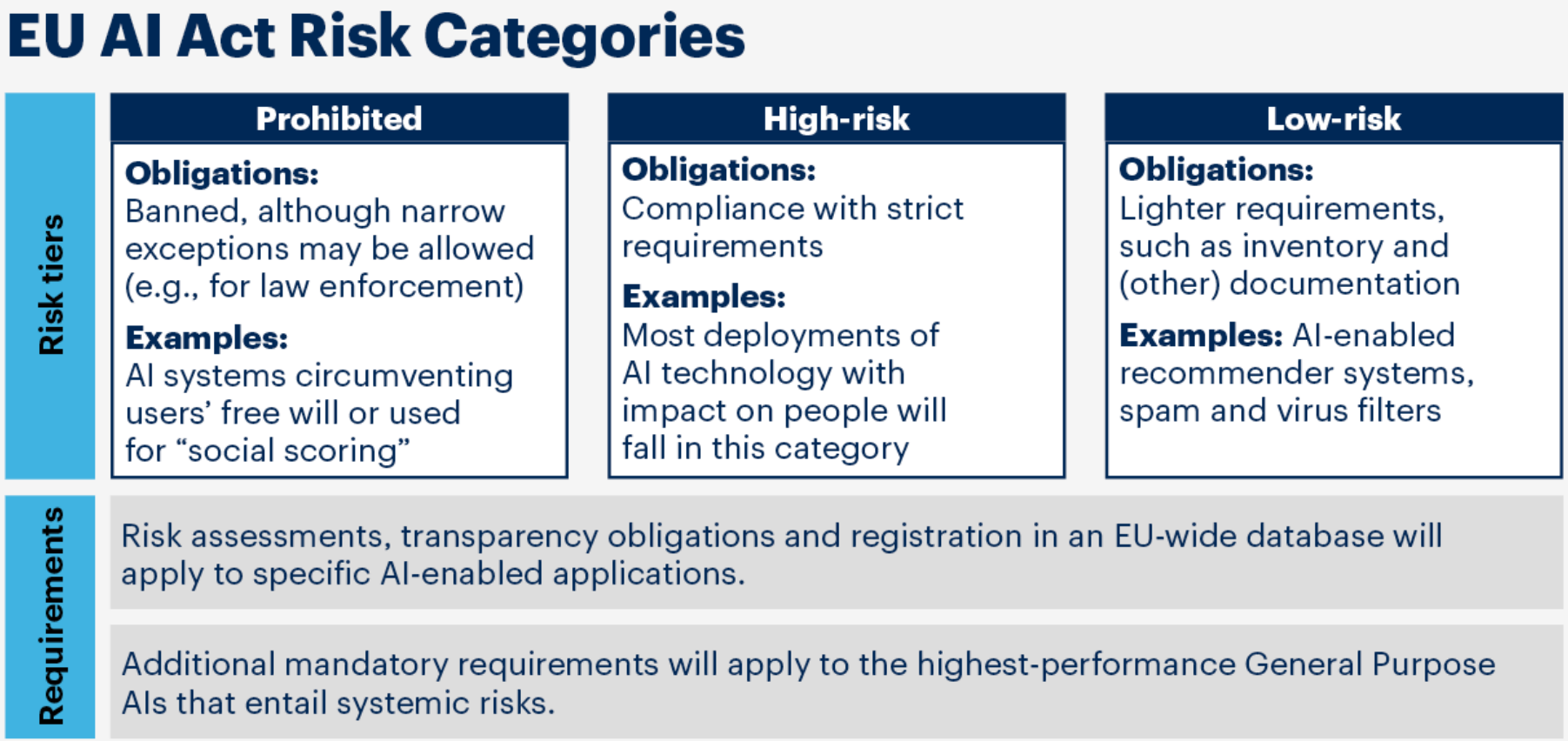
Aggregation bias arises when flawed assumptions about the population affect model definition. In many applications, the population of interest is heterogeneous and a single model is unlikely to suit all subgroups.

11.4 Regulation of AI at Different Levels in the European Union



Adapted from the Artificial Intelligence Act - Regulation (EU) 2024/1689.

11.4 EU AI Act Risk Categories



Adapted from the Artificial Intelligence Act - Regulation (EU) 2024/1689.

11.4 Implications for AI Development in the EU

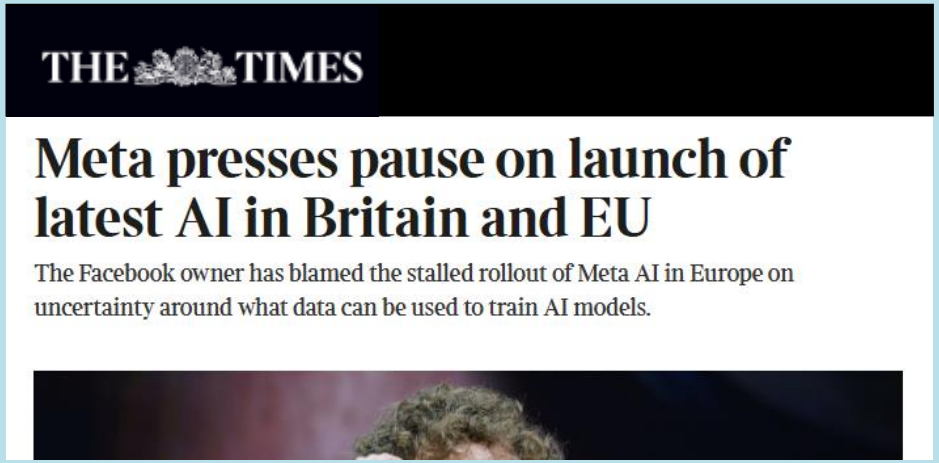
- Development costs rise due to mandatory compliance processes like data quality checks, risk assessments, and third-party evaluations
- The EU's stringent rules may place its companies at a disadvantage compared to less-regulated regions (e.g., US, China)
- Companies must maintain extensive records of training data, model design, and decision-making processes
- High fines and penalties: Non-compliance can lead to significant financial penalties (up to €30M or 6% of global annual turnover)!

Adapted from the Artificial Intelligence Act - Regulation (EU) 2024/1689.

11.4 The EU AI Act: Pioneering Leadership or Strategic Misstep?







11.4 Classroom Task

Your turn!

Task

Consider the following aspects, what are ethical problems and possible biases in AI usage and AI design.

- Automation of communication (e.g. Case Ashley Madison)
- Homework automation/solutions with ChatGPT
- Generation of fake image and videos (e.g. deep fakes)



Image source: ↗ Pixabay (2019) / ↗ CC0

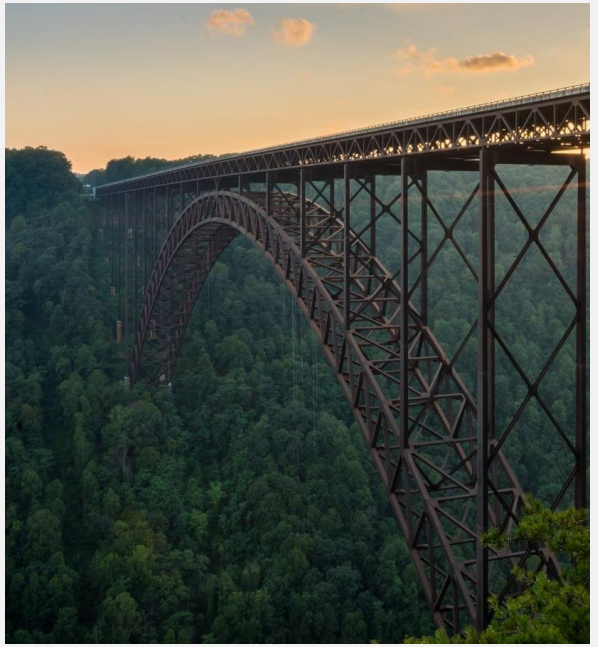
Outline

11 Building Productive AI-based Systems

- 11.1 The Challenge of Applied AI
- 11.2 Robotic Agents
- 11.3 Human-AI Interaction
- 11.4 AI Ethics and Governance
- 11.5 How to Continue your AI Journey

Exam Preparation and Course Repetitorium

Capstone Project / Case Challenge



► **What you will learn:**

- Why many AI-based Information systems and intelligent agents fail in practice and how to tackle common problems in implementing AI-based information systems and intelligent agents
- Outlook and Future steps if you are interested in an AI-job

► **Duration:**

- 135 min + FAQs

► **Relevant for Exam:**

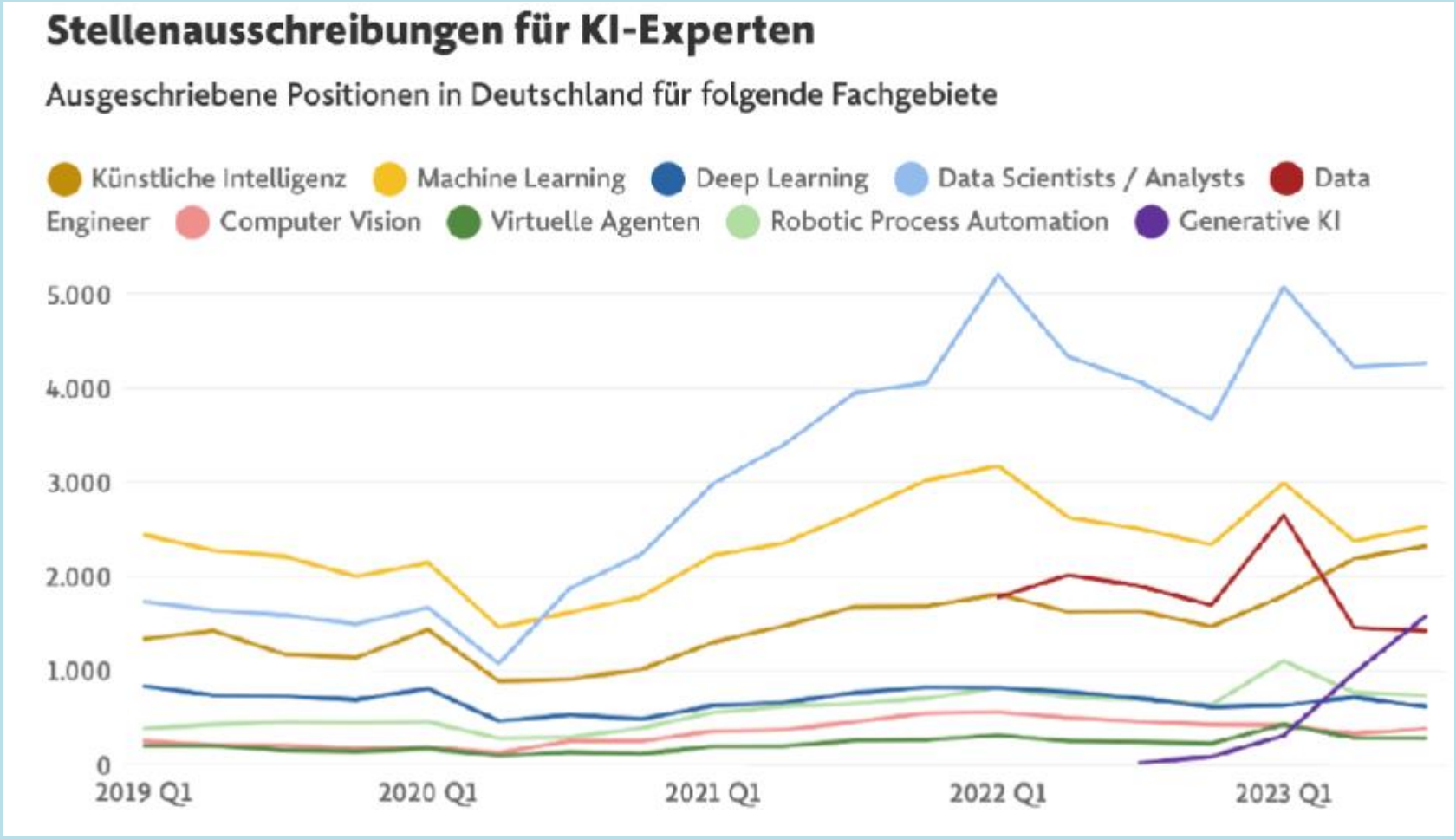
- 11.1-11.4

11.5 Start your AI Journey



Image source: ↗ Pixabay (2019) / ↗ CC0

11.5 How is the Job Situation?

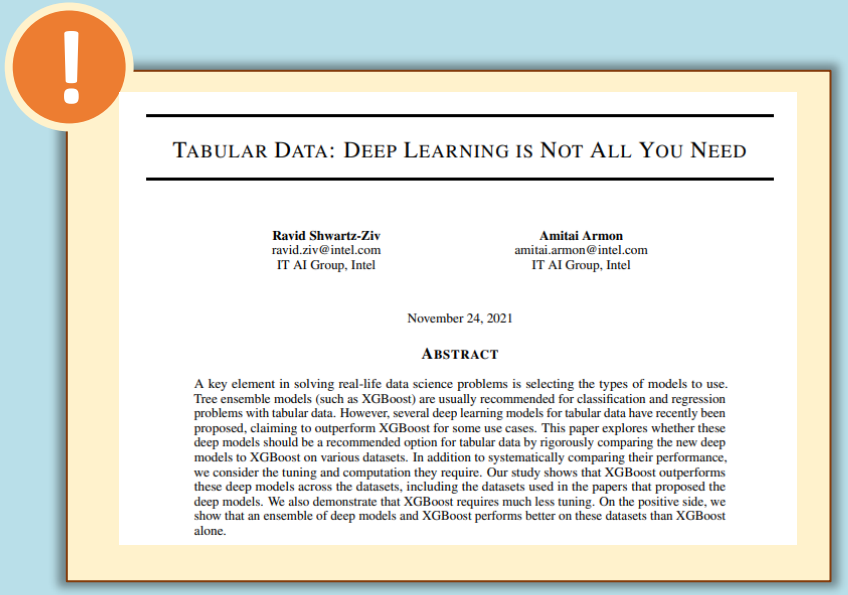




Adapted from FAZ PRO, D:Economy (2023), based on saco / Indec Gruppe 2023

11.5 Untold Truths about AI Jobs (personal opinion !)

- Big gap between theoretical vs. practical work (aka university vs. industry)
- There are many misleading learning resources
- A solid background in the foundations (e.g. linear algebra) is crucial
- Productization is Key
- Neural Networks are not the silver bullet
- Challenges Often Aren’t Technical
- Impact Over Skills List



TABULAR DATA: DEEP LEARNING IS NOT ALL YOU NEED

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November 24, 2021

ABSTRACT

A key element in solving real-life data science problems is selecting the types of models to use. Tree ensemble models (such as XGBoost) are usually recommended for classification and regression problems with tabular data. However, several deep learning models for tabular data have recently been proposed, claiming to outperform XGBoost for some use cases. This paper explores whether these deep models should be a recommended option for tabular data by rigorously comparing the new deep models to XGBoost on various datasets. In addition to systematically comparing their performance, we consider the tuning and computation they require. Our study shows that XGBoost outperforms these deep models across the datasets, including the datasets used in the papers that proposed the deep models. We also demonstrate that XGBoost requires much less tuning. On the positive side, we show that an ensemble of deep models and XGBoost performs better on these datasets than XGBoost alone.

11.5 Recap - Job Profiles in AI (Examples)

AI Specialist

- Professional responsible for AI related topics at companys
- Designing, developing and maintaining simple AI related solutions
- Product owner in AI-related projects

Data Engineer

- Models scalable database and data flow architectures
- Develops and improves the IT infrastructure on the hardware and software side
- Deals with topics such as IT Security , Data Security and Data Protection

Data Scientist

- AI-Expert responsible for collecting, analyzing and interpreting extremely large amounts of data
- The role is an offshoot of several traditional technical roles, including mathematician, scientist, statistician and computer professional

Robotics Scientist

- Engineer responsible for implementing intelligent robots
- Bridge between mechanical engineering, electrical engineering, computer science

Business Data Analyst / BI Developer

- Designing, developing and maintaining business intelligence solutions
- Crafting and executing queries upon request for data
- Presenting information through reports and visualization

Machine Learning Engineer

- Engineer responsible for implementing intelligent robots
- Bridge between mechanical engineering, electrical engineering, computer science

AI Research Scientist

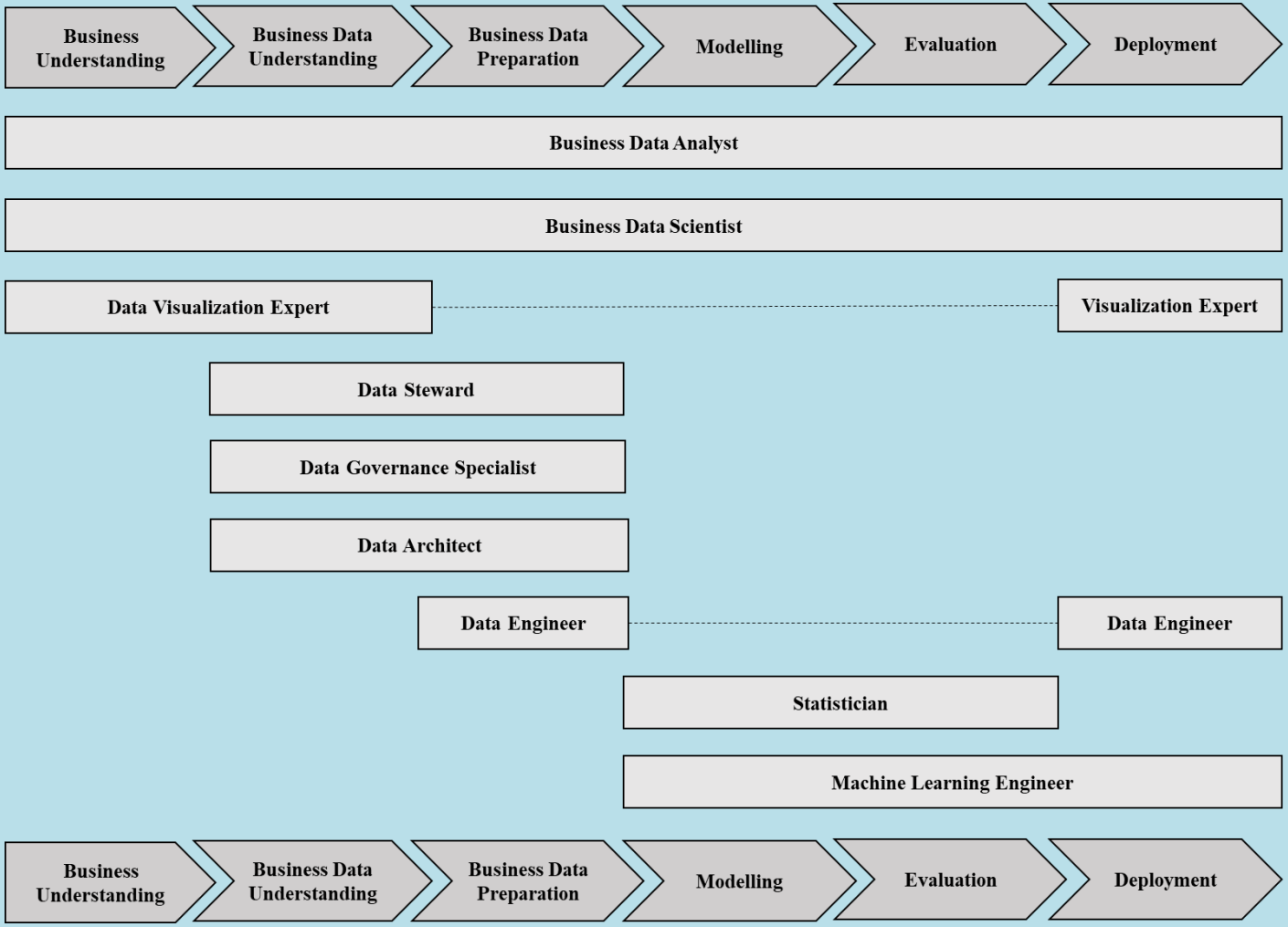
- Works mostly for universities or big companies
- Passionately drive and further advance innovations in the field of computer vision

AI Product Owner

- Defining the vision and strategy of AI-driven products
- Ensuring that the development team creates solutions that meet business goals and user needs.

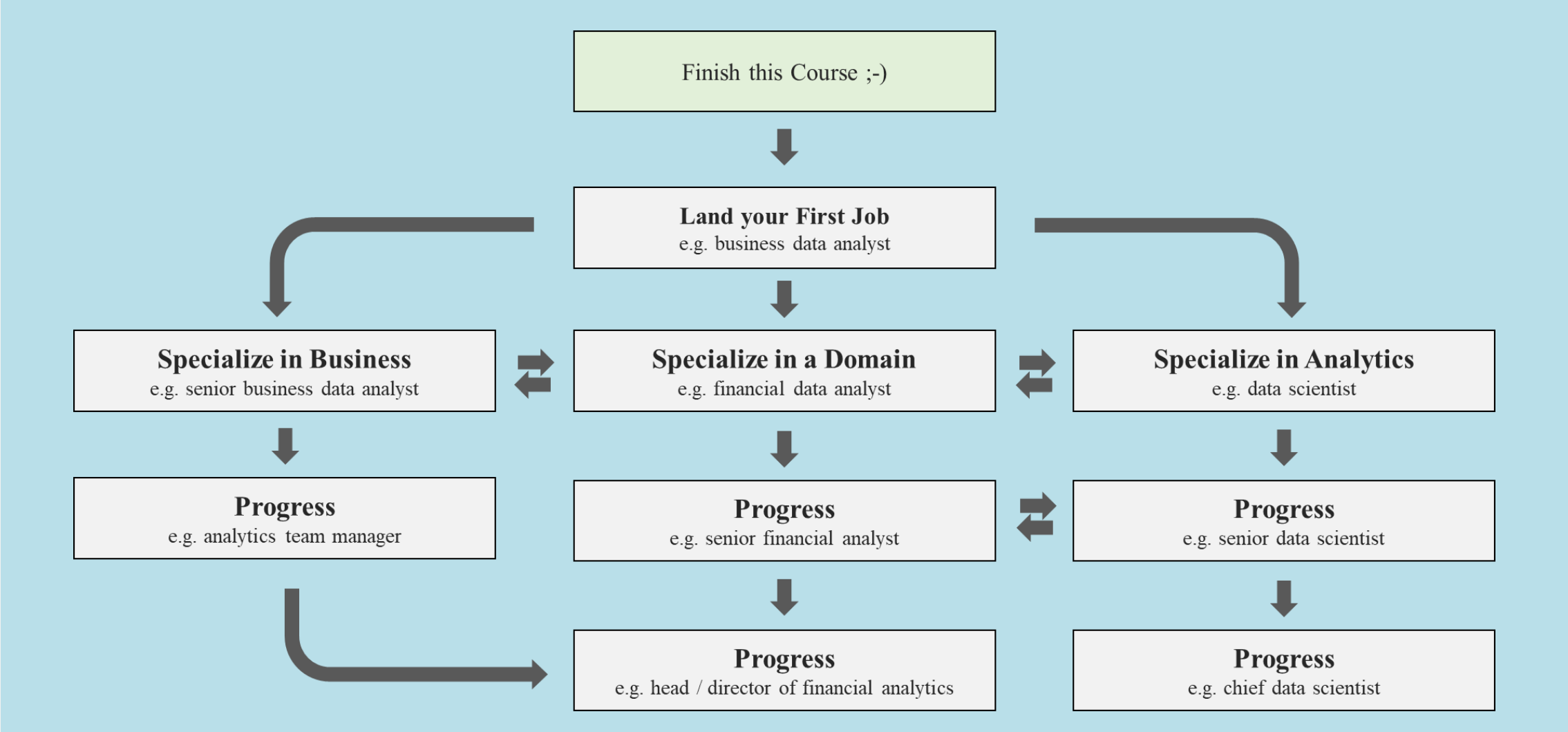
...

11.5 Possible Job Roles Across an AI Project



Adapted from Jung, D. (2024)

11.5 Possible Career Paths in the Industry



Adapted from Jung, D. (2024)

11.5 Getting the Right Experience – University Courses

I can recommend to take courses in the following subjects to widen your practical AI skillset:

Computer Science
- Data Management and Warehousing
- Software Engineering and Architecture
- Algorithms

Statistics and Math
- Oeconometrics
- Operations Research

Specialization in AI
- Natural Language Processing,
- Robotics,
- Machine Learning (!)
- …

Or Combine with other domains:
- Bio-Informatics
- Finance
- …

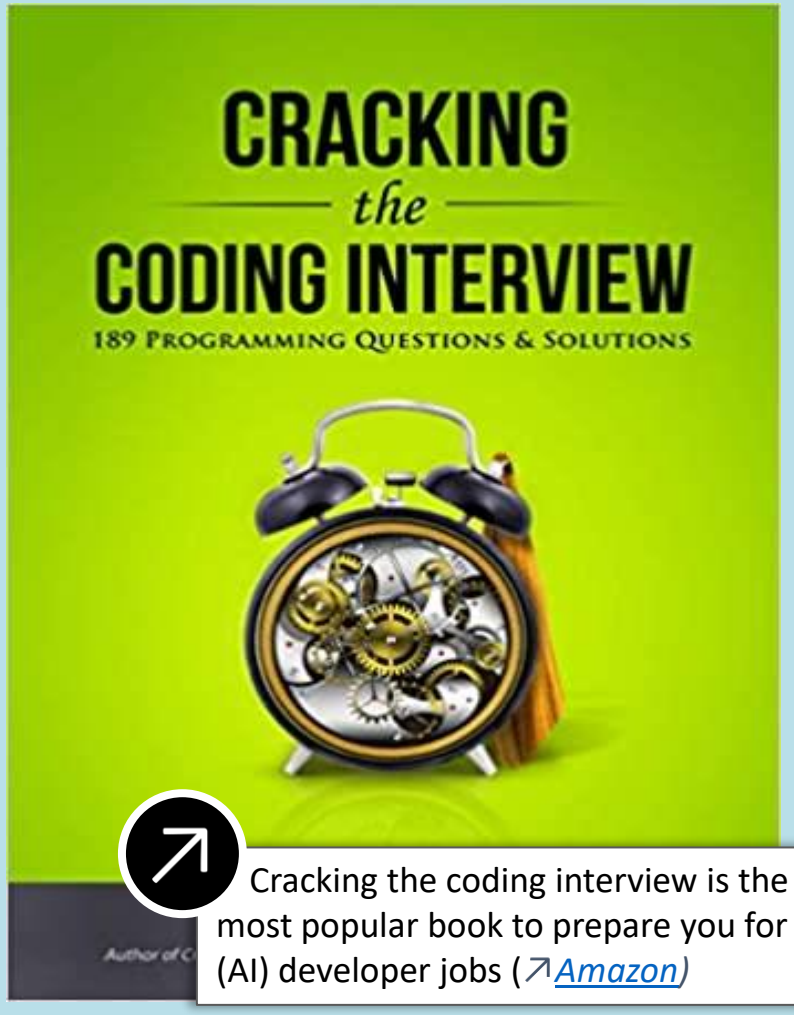


You have questions related to your curriculum? Just feel free to ask now or come to me after lecture!

11.5 Getting the Right Experience – PhD or Not

11.5 Preparing the Technical Interview

- If you apply for an AI job in big or in IT companies you will probably face a coding interview
- Coding interviews require some preparation (or many many years experience as AI developer)
- Best preparation is to apply for student jobs besides your studies



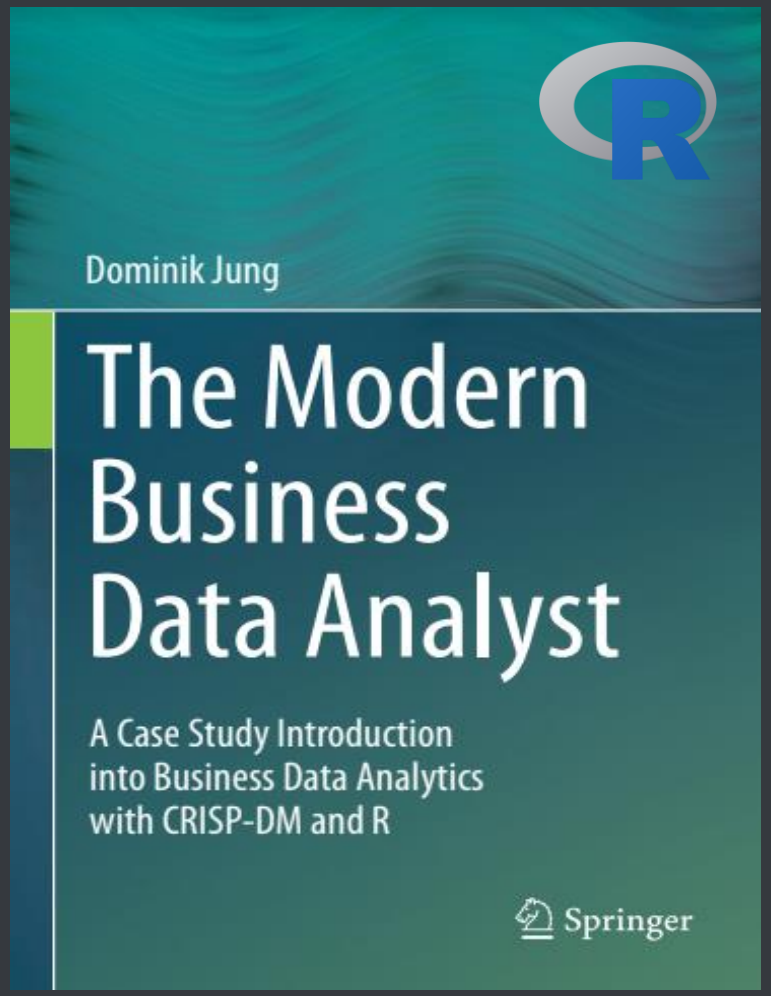
Cracking the coding interview is the most popular book to prepare you for (AI) developer jobs (↗*Amazon*)

11.5 Maintaining Your Competitive Advantage is Hard



- Half-life of AI knowledge (see e.g. DistBelief, Toolboxes, AutoML)
- AI tools will change and become more and more easy to use
- Better data will trump AI skills
- Do not rely soley on your AI Skills

11.5 Book Recommendation if You Want to Become Business Data Analyst



Contents:

- Starting-point for data science beginners and data analysts
- Hands-on examples and real business problems with R
- Accompanying course website and business cases to practice

Accompanying course:



Business Data Analytics and Decision Support with R
↗ course materials

↗ The Modern Business Data Analyst (↗Amazon | ↗ Springer

Your next project:
AI Capstone Challenge



Example: 2021 Capstone with Warranty and Goodwill teams @Porsche AG

11. Exercises

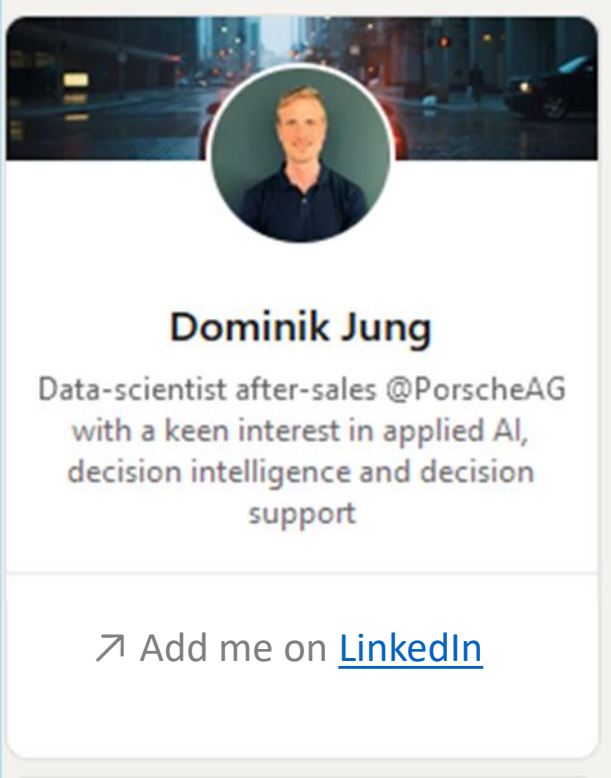
Workbook Exercises

- Please read the chapter 25 from Rusell, S., & Norvig, P. (2016) to understand how to build and deploy agents and robotic AI-based systems. Then continute with chapters 26 to 27 and reflect the pros and cons of AI for i) your life, ii) everydays life iii) application in industry iv) civilization. Then work through the exercises of the chapters to prepare for the exam.

Coding Exercises

- Due to capstone project or case challenge, there will be no coding exercises in this chapter

10. Feel Free to Add me on LinkedIn



- If you want to stay in contact or if have further job-related questions, you can add me on LinkedIn. I try to answer most messages on my weekends.
- Additionally, I try continuously to improve and update the course material on git, hence if you have any ideas, wishes, or questions please feel free to write me.


You will do well in your future endeavors!



Microsoft AI ft. Common
https://www.youtube.com/watch?v=9tucY7Jhhs4

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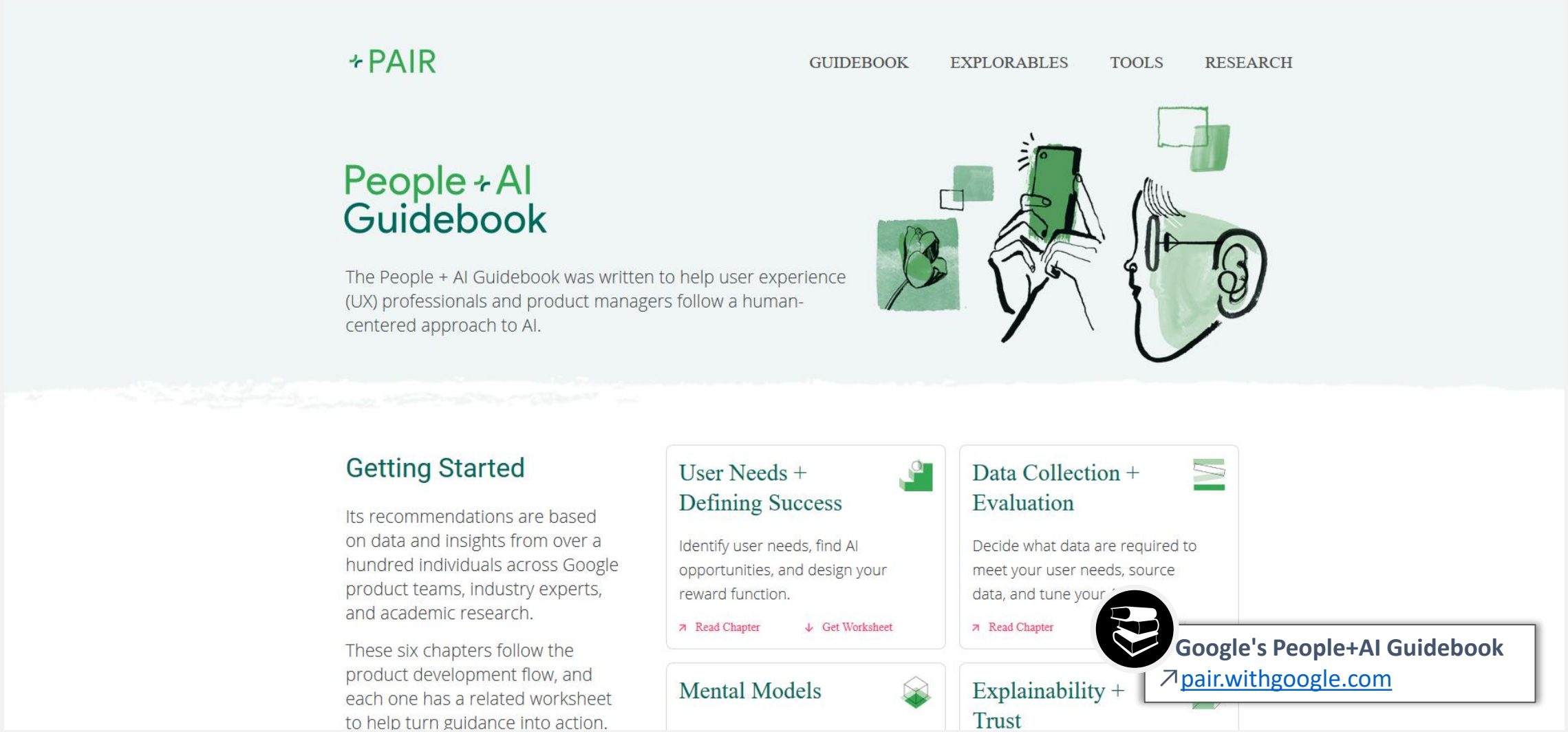
Images

All images that were not marked other ways are made by myself, or licensed ↗CC0 from ↗Pixabay.

Further reading

- I strongly recommend to take a look at Design Blog from Google (↗ Google Design), where the Designers from Google share their knowledge and best practices. Further intersting tutorials and best practices you find on Google Developpers Guide Archive (↗ Google Developers). Both links are a must-read for information systems developer!
- The famous republica study raising very relevant questions about the usage of the COMPASS algorithm in the US is online available at: ↗ www.propublica.org
- For an overview of political biases in LLM see e.g. Feng, S et al (2023).

11. Google's People+AI Guidebook (Google Research, Google Design)



11. Glossary

AI Bias	Systematic errors in AI systems that result in unfair or prejudiced outcomes due to biases in training data or algorithm design
CRISP-DM	A widely used framework for developing data science projects, comprising stages like business understanding, data understanding, data preparation, modeling, evaluation, and deployment.
Explainable AI	*Explainable AI (XAI) is artificial intelligence in which the results of the solution can be understood by humans. It contrasts with the concept of the "black box" in machine learning*
EU AI Act	A regulatory framework by the European Union aimed at categorizing AI systems based on risk levels and imposing compliance measures, with significant implications for businesses operating in the EU.
Lifecycle Approach	A structured methodology for AI project development, spanning the inception of an AI use case through to deployment and maintenance.
XAI / Human-AI Interaction	Guidelines and practices for designing AI systems that effectively interact with users, covering aspects like user trust, error handling, and adaptability over time.

Bucket List - US National Parks

1872 Yellowstone
1890 Yosemite (CA SP 1864)
Sequoia
1899 Mount Rainier
1902 Crater Lake
1903 Wind Cave
1906 Mesa Verde
1910 Glacier
1915 Rocky Mountain
1916 Lassen Volcanic (Cinder Cone NM & Lassen Peak NM 1907)
Hawaii Volcanoes (originally part of Hawaii NP)
Haleakala (originally part of Hawaii NP)
1917 Denali (originally Mt McKinley NP, renamed in 1980)
1919 Grand Canyon (NM 1908)
Zion (Mukuntuweap NM 1909)
Acadia (Sieur de Monts NM 1916, originally Lafayette NP, renamed in 1929)
1921 Hot Springs (Reservation 1832)
1926 Shenandoah (established 1935)
1928 Bryce Canyon (NM 1923)
1929 Grand Teton
1930 Carlsbad Caverns (1923)
1934 Everglades (established 1947)
Great Smoky Mountains (established 1940)
1938 Olympic (Mt Olympus NM 1909)
1940 Kings Canyon
Isle Royale
1941 Mammoth Cave
1944 Big Bend
1956 Virgin Islands
1962 Petrified Forest (NM 1906)
1964 Canyonlands
1966 Guadalupe Mountains (established 1972)
1968 North Cascades
Redwood
1971 Capitol Reef (NM 1937)
Voyageurs (established 1975)
Arches (NM 1929)
1978 Theodore Roosevelt (National Memorial Park 1947)
Badlands (NM 1929)
1980 Channel Islands (NM 1938)
Biscayne (NM 1968)
Katmai (NM 1918)
Glacier Bay (NM 1925)
Gates of the Artic (NM 1978)
Kenai Fjords (NM 1978)
Kobuk Valley (NM 1978)
Lake Clark (NM 1978)
Wrangell St. Elias (NM 1978)
1986 Great Basin (Lehman Caves NM 1922)
1988 National Park of American Samoa (established 1993)
1992 Dry Tortugas (Fort Jefferson NM 1935)
1994 Death Valley (NM 1933)
Saguaro (NM 1933)
Joshua Tree (NM 1936)
1999 Black Canyon of the Gunnison (NM 1933)
2000 Cuyahoga Valley (National Recreation Area 1974)
2003 Congaree (Congaree Swamp NM 1976)
2004 Great Sand Dunes (NM 1932)
2013 Pinnacles (NM 1908)
2018 Gateway Arch (Jefferson National Expansion Memorial 1935)
2019 Indiana Dunes (National Lakeshore 1966)
White Sands (NM 1933)
2020 New River Gorge (National River 1978